

Anticipating demand in carbon-aware scheduling across data centres: guarantees and fair carbon attribution

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Abstract

Shifting deferrable jobs to cleaner hours and locations cuts data-centre emissions, but online schedulers act before future grid conditions and jobs are known. We formulate multi-site, deadline-constrained scheduling of bag-of-tasks jobs as a minimum-cost flow problem and propose CARMA (Carbon-aware Anticipatory Receding-horizon Multi-site Allocation), which adds “ghost jobs” for expected arrivals to each hourly plan. We prove that uncertainty about demand alone forces any online algorithm, even one with perfect carbon forecasts, to a competitive ratio of $\Omega(\sqrt{\theta})$, where θ is the ratio of the highest to the lowest unit emission, and myopic planning to at least $(1 + \theta)/2$. Given backstop capacity, CARMA never misses a deadline under any predictions, is optimal under exact predictions and degrades linearly with prediction error. Dual prices of the pooled problem yield a carbon attribution among sites that lies in the core. On 2024 regional grid data for Great Britain, CARMA comes within 8.0–9.8% of the clairvoyant optimum, against 15–16% for the best myopic, protection-level or risk-averse baseline and 19–23% for plain model-predictive control, and cuts attributional emissions by 36–68% relative to carbon-agnostic operation. On a real cluster trace with a demand profile learned from different days, its gap is 20.7% against 26.7–36.1%. Anticipating demand is worth more than perfect carbon foresight. Under stylised marginal-emission accounting, achievable reductions shrink to 14% and CARMA’s edge becomes small, so physical savings depend on curtailed low-carbon supply. Origin- and host-based carbon accounting violate coalitional stability in every week, whereas the dual attribution never does in hindsight.

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1. Introduction

Data centres are among the fastest-growing electricity consumers. Efficiency gains kept their energy use nearly flat through the 2010s even as computing output grew several-fold [1, 2]. That buffer is now thinning, and information and communication technology accounts for an estimated 2.1–3.9% of global greenhouse-gas emissions [3]. Large machine-learning workloads have drawn particular attention [4, 5, 6]. The carbon intensity of grid electricity varies by a factor of ten or more across regions and hours, driven by the output of wind, solar and dispatchable plant [7]. Operators can therefore cut emissions without new hardware by moving flexible work to cleaner places and times. Such *carbon-aware computing* [8, 9] is an operations problem: capacity is finite, deadlines bind, and both the grid and the job stream are uncertain.

Batch jobs such as model training, analytics, rendering and backups are a natural target, because they tolerate some delay. Previous work has shown the potential of temporal shifting (running later) [10, 11, 12] and of spatial shifting (running elsewhere) [13, 14]. It has also shown their limits. Sukprasert et al. [15] found that practical reductions from spatio-temporal shifting fall well short of idealised bounds. Most methods are evaluated with a scheduler that reacts to the jobs it has already seen, using point forecasts of carbon intensity. Three questions have received less attention. First, how much is lost because an online scheduler does not know which jobs will arrive next, and can any scheduler guarantee good performance in the worst case? Second, does it pay to price the uncertainty of carbon-intensity forecasts into scheduling decisions? Third, when several sites pool their capacity, how should the resulting emissions be attributed so that every participant is better off than on its own?

This paper answers these questions with a new algorithm, provable guarantees and a forecasting model, evaluated on three years of real regional grid data for Great Britain. It makes five contributions.

1. **Formulation.** We state multi-site, deadline-constrained, width-limited scheduling of deferrable bag-of-tasks jobs, including the emissions of

data transfer, as a minimum-cost flow problem (Proposition 1). Offline clairvoyant optima and every online planning step can therefore be computed exactly and quickly by network simplex. We also prove that myopic receding-horizon scheduling can be arbitrarily worse than the clairvoyant optimum even with perfect carbon forecasts (Proposition 2).

2. **Algorithm.** We propose CARMA (Carbon-aware Anticipatory Receding-horizon Multi-site Allocation). At every hour, CARMA adds *ghost jobs* to its planning network. These represent expected arrivals, learned from the operator’s history, so the planner reserves scarce low-carbon capacity for work that has not yet arrived. The planning problem stays a single min-cost flow and decisions take tens of milliseconds.
3. **Guarantees.** We prove that uncertainty about demand alone forces every online algorithm, even one with perfect carbon forecasts, to a competitive ratio of $\Omega(\sqrt{\theta})$, about $\sqrt{\theta}/2$ for large θ , where θ is the ratio of the highest to the lowest unit emission (Theorem 1). We show that, given a backstop resource, CARMA never misses a deadline whatever its predictions, so it inherits the θ -competitiveness that every on-time policy has. It is also optimal when its predictions are exact, and its excess emissions grow at most linearly in the prediction errors (Proposition 3 and Theorem 2). We also show that the dual prices of the pooled scheduling network give a carbon attribution among sites that lies in the core of the pooling game, so no group of sites gains by leaving (Theorem 3).
4. **Forecasting model.** We develop a probabilistic 1–24-hour carbon-intensity forecaster. It consists of direct gradient-boosted models with horizon-wise normalised conformal calibration and adaptive conformal inference, which holds coverage under the strong decarbonisation drift seen in 2024. Its calibrated quantiles let us test risk-adjusted carbon pricing inside the scheduler.
5. **Evidence.** Across 52 out-of-sample weeks with three workload traces each, three load levels, two operating regimes and a trace-driven workload from the Alibaba 2018 cluster trace, we compare CARMA with six implementable online baselines, including a revenue-management protection-level policy, with clairvoyant and ablated variants, and with the clairvoyant oracle. The results separate the value of anticipating demand from the value of better carbon forecasts, show when each matters, confirm the graceful degradation predicted by the theory, and

71 show that common location-based carbon accounting violates coalitional
 72 stability in every test week, whereas the dual attribution satisfies it
 73 exactly for the hindsight benchmark and in over 99% of the checks for
 74 CARMA’s realised emissions.

75 The results carry an operational message. Once spatial shifting is allowed,
 76 the dominant loss comes from demand myopia, not from forecast error.
 77 Anticipating demand cuts the gap to the clairvoyant optimum by more than
 78 half in the synthetic experiments, and by 43% on a real cluster trace whose
 79 demand profile is learned from different days. Perfect carbon foresight or
 80 risk-averse carbon pricing adds little by comparison. Without migration,
 81 temporal shifting alone achieves much smaller reductions, and near saturation
 82 it trades emissions against lateness. We also show the limits of the evidence.
 83 The headline reductions are attributional. Under a stylised marginal-emission
 84 accounting within the single GB market, most of them are reallocations,
 85 and physical savings depend on low-carbon supply that would otherwise be
 86 curtailed. The rest of the paper is organised as follows. Section 2 reviews
 87 related work. Section 3 presents the model, the forecaster, CARMA and
 88 the experimental design. Section 4 establishes the theoretical guarantees,
 89 with proofs in Appendix A. Section 5 reports results, Section 6 discusses
 90 implications and limitations, and Section 7 concludes.

91 2. Related work

92 *Carbon-aware workload shifting.* Early green data-centre schedulers matched
 93 deferrable jobs to on-site solar production, for example GreenSlot [16] and
 94 GreenHadoop [11]. Other work exploited electricity-price and carbon differ-
 95 ences across sites through geographical load balancing [17, 13] and dynamic
 96 right-sizing [18]. With public carbon-intensity feeds, attention moved to grid
 97 emissions. Wiesner et al. [10] quantified the reductions from temporal shift-
 98 ing in several regions, including Great Britain, and proposed forecast-based
 99 lowest-window scheduling. Google’s carbon-intelligent compute system sets
 100 next-day virtual capacity curves from carbon forecasts and forecasts of the
 101 fleet’s own demand [8]. It is the closest industrial precedent for using demand
 102 forecasts, but it shapes hourly virtual capacity curves per cluster rather
 103 than planning individual jobs across sites. Cucumber admits delay-tolerant
 104 work only when forecast renewable supply allows it [19], and Caribou shifts
 105 serverless functions between regions [20]. CarbonScaler varies the allocation

106 of elastic jobs with carbon intensity [12]. Ecovisor exposes the energy system
 107 to applications [21]. Carbon Explorer weighs scheduling against investment
 108 in renewables and storage [22]. Load migration between data centres can
 109 absorb curtailed renewable generation [14], and data-centre demand response
 110 has been surveyed in [23]. Sukprasert et al. [15] bound the potential of spatio-
 111 temporal shifting across 123 regions and find that simple policies capture
 112 most of the achievable benefit. Our study complements theirs by identifying
 113 *which* information limits online schedulers when capacity at clean sites is
 114 scarce.

115 *Online algorithms and guarantees.* Worst-case analysis of carbon-aware
 116 scheduling builds on online search and one-way trading [24], for which
 117 threshold policies achieve optimal competitive ratios. Lechowicz et al.
 118 derived optimal double-threshold algorithms for the online pause-and-resume
 119 problem [25]. With CarbonClipper, they also derived learning-augmented
 120 algorithms with an optimal consistency–robustness trade-off for spatiotem-
 121 poral allocation of a single deadline-constrained workload over a network
 122 of data centres [26]. These works model uncertainty about future carbon
 123 *prices* for one job. We study many jobs that compete for finite low-carbon
 124 capacity, and we show that uncertainty about *demand* creates a distinct
 125 lower bound that carbon forecasts cannot remove. Our guarantees follow
 126 the consistency–robustness paradigm of algorithms with predictions [27].
 127 CARMA’s re-solving of a deterministic plan with expected future demand
 128 is an instance of the certainty-equivalent re-solving heuristics of network
 129 revenue management [28]. Our contribution is not the device itself but its
 130 formulation as a single exact flow problem for carbon-aware scheduling, the
 131 guarantees and the empirical attribution of value. Our fairness result draws
 132 on cooperative game theory: linear production games and flow games have
 133 non-empty cores that can be computed from dual prices [29, 30].

134 *Energy-aware scheduling in operations.* Energy-efficient scheduling is a mature
 135 topic in manufacturing operations [31], and green scheduling models have ap-
 136 peared in this journal, for example for parallel machines with job splitting [32]
 137 and for cloud resource allocation [33]. Most such models are deterministic and
 138 solved offline, often with metaheuristics. We instead exploit exact network-
 139 flow structure and embed it in an online lookahead policy. Network-flow
 140 formulations of preemptive scheduling with release dates and deadlines go
 141 back to Horn [34] and Federgruen and Groenevelt [35]. Proposition 1 extends

142 this classical construction to multiple sites, job widths, time-varying emission
 143 costs and transfer emissions [36, 37]. In the taxonomy of Powell [38], CARMA
 144 is a deterministic lookahead policy with a modified (anticipatory) model, in
 145 the spirit of model predictive control [39, 40].

146 *Carbon-intensity forecasting.* Forecasts of grid carbon intensity use statistical
 147 decomposition [41], machine learning with many exogenous drivers [42], day-
 148 ahead regression for building demand response [43], and hierarchical models of
 149 the generation mix, as in CarbonCast [44]. For Great Britain, NESO publishes
 150 regional forecasts through its Carbon Intensity API [45]. These are average
 151 (attributional) intensities. Marginal emission factors, which describe the plant
 152 that responds to a change in load, can differ substantially from averages and are
 153 the relevant measure for the system-level effect of load shifting [46, 47]. These
 154 studies focus on point accuracy. For decisions, calibrated uncertainty matters
 155 as well. Conformal prediction [48, 49, 50] provides distribution-free intervals
 156 and has been extended to quantile regression [51], to multi-horizon [52] and
 157 dynamic time series [53], and to arbitrary distribution shift through adaptive
 158 conformal inference [54]. We use these tools to build a calibrated forecaster
 159 and to test, in a controlled way, whether risk-adjusted carbon costs improve
 160 scheduling. The question is motivated by the optimizer’s curse [55] and by the
 161 robust-optimisation view that point-estimate solutions can be fragile [56, 57].

162 3. Material and methods

163 3.1. Problem setting

164 We consider an operator that runs a set $\mathcal{S}$ of data-centre sites connected
 165 by a wide-area network. Time is divided into hourly slots $t = 0, 1, 2, \dots$
 166 Deferrable batch jobs arrive online. Job j is released at the start of hour a_j
 167 at origin site $o_j \in \mathcal{S}$. It carries w_j server-hours of work made of independent
 168 tasks (a bag of tasks), can use at most m_j servers concurrently, and must
 169 finish before the start of hour $d_j > a_j$. Work may be paused and resumed,
 170 and any part of it may run at a site other than its origin. Remote execution
 171 requires moving input data. At site s and hour t , $K_s(t)$ servers are free for
 172 batch work once the interactive load has been served.

173 Let $I_s(t)$ be the average carbon intensity of the electricity supplied to site
 174 s during hour t (gCO₂/kWh). Executing one server-hour of job j at site s
 175 during hour t emits

$$c_{o_j s}(t) = e \text{PUE}_s I_s(t) + \mathbf{1}[s \neq o_j] \delta \eta \frac{1}{2} (I_{o_j}(t) + I_s(t)), \quad (1)$$

176 where e is the IT energy per server-hour, PUE_s is the power usage effectiveness
 177 of site s , δ is the data volume moved per migrated server-hour and η is the
 178 energy intensity of network transfer (kWh/GB). The transfer term charges
 179 network energy at the mean intensity of the two endpoints. With decision
 180 variables $x_{js}(t) \geq 0$ (server-hours of job j run at site s in hour t), the offline
 181 version of the problem is the linear programme

$$\min_{x, u \geq 0} \sum_j \sum_s \sum_{t=a_j}^{d_j-1} c_{ojs}(t) x_{js}(t) + \sum_j \Pi u_j \quad (2)$$

$$\text{s.t.} \quad \sum_s \sum_{t=a_j}^{d_j-1} x_{js}(t) + u_j = w_j \quad \forall j, \quad (3)$$

$$\sum_s x_{js}(t) \leq m_j \quad \forall j, t \in [a_j, d_j), \quad (4)$$

$$\sum_{j: a_j \leq t < d_j} x_{js}(t) \leq K_s(t) \quad \forall s, t, \quad (5)$$

182 where u_j is work left unfinished at the deadline and Π is a lateness penalty
 183 much larger than any emission cost. When migration is not allowed, $x_{js}(t) = 0$
 184 for $s \neq o_j$. With complete knowledge of arrivals and intensities, the optimum
 185 of (2)–(5) is a lower bound on the emissions of every schedule that serves
 186 all jobs by their deadlines, and we use it as the *clairvoyant oracle*. When
 187 emissions are evaluated only for a subset of jobs (Section 3.5), the oracle still
 188 serves every job but charges emissions only for the evaluated ones, so that it
 189 remains a lower bound on the evaluated emissions. An online scheduler has
 190 to fix $x_{.s}(t)$ at the start of hour t . At that point it knows the jobs released so
 191 far, the current intensities $I_s(t)$ and nothing more than forecasts of $I_s(t+h)$
 192 and of future arrivals.

193 **Proposition 1** (Network-flow structure). *Problem (2)–(5) is a minimum-cost*
 194 *flow problem on a network with a source, one node per job, one node per (job,*
 195 *hour) pair, one node per (hour, site) pair and a sink. Arc capacities are w_j*
 196 *(source $\rightarrow j$), m_j ($j \rightarrow (j, t)$) and $K_s(t)$ ($(t, s) \rightarrow \text{sink}$). The only arcs with a*
 197 *cost are $(j, t) \rightarrow (t, s)$, priced at $c_{ojs}(t)$, and a slack arc $j \rightarrow \text{sink}$ priced at Π .*

198 *Proof.* Conservation at the job node is Eq. (3). The capacity of arc $j \rightarrow (j, t)$
 199 is Eq. (4), conservation at (j, t) splits that hour's work across sites, and
 200 the capacity of $(t, s) \rightarrow \text{sink}$ is Eq. (5). Conversely, every feasible flow
 201 defines a feasible (x, u) with the same cost. The constraint matrix of the flow
 202 formulation is a node-arc incidence matrix, in which every column has exactly
 203 one +1 and one -1, so it is totally unimodular. $\square$

204 Proposition 1 has a practical consequence. The oracle and every plan-
 205 ning problem solved online (Section 3.3) can be solved exactly by network
 206 simplex [36], and integral data (work quantised to 0.01 server-hours) give
 207 integral optimal flows. We use the OR-Tools implementation [58]. On the
 208 weekly clairvoyant problems (1,100–2,800 jobs), network simplex returned
 209 the same objective as a general-purpose LP solver (HiGHS) to within a rel-
 210 ative 5×10^{-5} and was 3–8 times faster (0.2–0.9 s against 0.5–4.4 s; script
 211 `bench_solvers.py`).

212 3.2. Probabilistic carbon-intensity forecasting

213 *Point model.* For each horizon $h = 1, \dots, H$ ($H = 24$ h) we fit a direct
 214 gradient-boosted regression tree model [59] that predicts the change $I_s(t +$
 215 $h) - I_s(t)$ from information available at issue time t . Direct per-horizon
 216 models avoid the error accumulation of recursive strategies [60]. One model
 217 per horizon is pooled across sites, with the site as a categorical feature. The
 218 features are the current intensity; differences between the current value and
 219 lags of 1, 2, 3, 6, 12 and 23 hours; the value observed at the same hour of
 220 the target on the previous day and in the previous week; a 24-hour rolling
 221 mean and mean absolute change (volatility); the national (GB) intensity; the
 222 regional wind, solar and gas shares; and calendar encodings of the target
 223 time (hour of day, day of week, annual sine and cosine). Models are fitted
 224 with absolute-error loss, so they estimate the conditional median. We use
 225 scikit-learn’s histogram gradient boosting [61] with 400 iterations, learning
 226 rate 0.05, 31 leaves and at least 50 samples per leaf. The models use only
 227 information available when the forecast is issued, and no weather forecasts.
 228 This is deliberate: the goal is a forecaster any operator can build from a
 229 public intensity feed.

230 *Conformal upper quantiles.* A point forecast $\hat{I}_s(t + h | t)$ is turned into cali-
 231 brated upper quantiles by split conformal prediction [48, 49] with normalised
 232 scores. On a calibration window disjoint from training, we compute for each
 233 site s and horizon h

$$r_{s,h}(t) = \frac{I_s(t + h) - \hat{I}_s(t + h | t)}{\sigma_s(t) + \epsilon}, \quad (6)$$

234 where $\sigma_s(t)$ is the 24-hour mean absolute hourly change observed at issue
 235 time and $\epsilon = 5$ gCO₂/kWh. For a target level β , the upper forecast is
 236 $U_s^\beta(t + h | t) = \hat{I}_s(t + h | t) + \hat{q}_{s,h}(\beta_{s,h}(t))(\sigma_s(t) + \epsilon)$, where $\hat{q}_{s,h}(\cdot)$ is the

empirical quantile function of the calibration scores. Grid carbon intensity is not exchangeable over time: 2024 was markedly cleaner than 2022–2023 (Fig. 3). We therefore update the working level online by adaptive conformal inference (ACI) [54]. Once $I_s(t+h)$ is observed,

$$\alpha_{s,h} \leftarrow \alpha_{s,h} + \gamma((1-\beta) - \mathbf{1}[I_s(t+h) > U_s^\beta(t+h | t)]), \quad \beta_{s,h}(t) = 1 - \alpha_{s,h}, \quad (7)$$

with step size $\gamma = 0.005$. By the ACI result of [54], the long-run frequency with which realised intensities exceed U^β converges to $1 - \beta$ for any data sequence, with no distributional assumption. This matters for the scheduler: the level β becomes an interpretable risk dial, and it stays valid under drift. With $\beta = 0.5$ the procedure also removes level shifts in the point model (a recalibrated median).

3.3. CARMA: carbon-aware anticipatory receding-horizon multi-site allocation

A standard receding-horizon (model predictive control, MPC) scheduler [39] solves, at every hour t , the flow problem of Proposition 1 restricted to pending jobs and the window $[t, t+H)$. Future intensities are replaced by forecasts, and only the first hour of the plan is executed. Such a planner is *myopic about demand*: it sees only jobs that have already arrived. The next result shows that this can be arbitrarily costly even when carbon intensities are known exactly.

Proposition 2 (Myopia can be unboundedly costly). *Even with perfect intensity forecasts, myopic receding-horizon scheduling has no finite competitive ratio with respect to the clairvoyant oracle.*

Proof. Take a clean site A with one server and a dirty site B with ample capacity. Let $c_A(0) = \varepsilon$, $c_A(1) = 0$ and $c_B(0) = c_B(1) = c$, and ignore transfer costs. A flexible job ($w = 1$, $m = 1$) arrives at $t = 0$ with deadline $d = 2$. An urgent job ($w = 1$) arrives at $t = 1$ with $d = 2$. At $t = 0$ the myopic planner sees only the flexible job and defers it to A at $t = 1$ (cost $0 < \varepsilon$). At $t = 1$ both jobs must run in the same hour, one of them at B , so the total cost is c . The oracle runs the flexible job at $t = 0$ and the urgent job at $t = 1$, both at A , for a total of ε . The ratio c/ε is unbounded as $\varepsilon \rightarrow 0$. $\square$

The mechanism in Proposition 2 is common in practice. Low-carbon capacity is scarce, and a myopic planner spends it on flexible work that could have waited or run elsewhere, leaving urgent jobs that arrive later to run on carbon-intensive capacity. CARMA addresses it by anticipating demand inside the same flow problem, and optionally prices forecast risk.

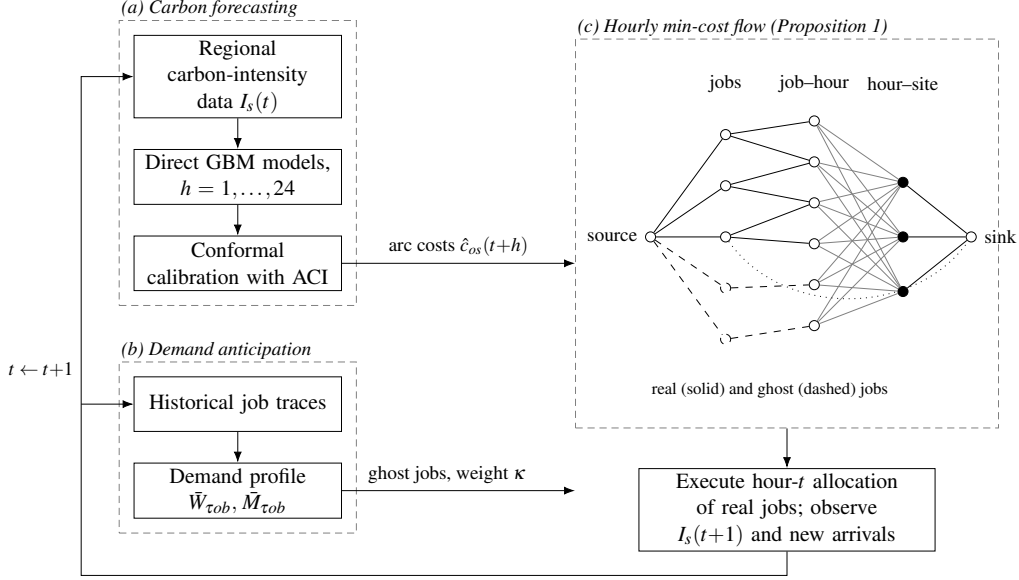


Figure 1: The CARMA decision loop. (a) Probabilistic carbon-intensity forecasts price the job-hour to hour-site arcs. (b) A demand profile learned from past traces generates ghost jobs for expected arrivals. (c) Real jobs (solid) and ghost jobs (dashed) share one min-cost flow network, whose dotted slack arc carries unfinished work to the sink at the lateness penalty. The network is solved exactly every hour, and only the first-hour allocation of real jobs is executed.

271 *Anticipated demand (ghost jobs)*. From the operator's own history we estimate,
 272 for every hour of the week τ , origin o and window class b , the expected work
 273 $\bar{W}_{\tau ob}$ and the expected total width $\bar{M}_{\tau ob}$ of arriving jobs. Window classes are
 274 the bins $\mathcal{B} = \{1, 2, 4, 6, 9, 13, 18, 24\}$ hours. A job is assigned to the largest
 275 bin not exceeding its window, so ghost jobs are never more flexible than the
 276 jobs they represent. Because binning shortens windows, the width of each
 277 ghost job is raised where necessary to at least its work divided by its window,
 278 which makes every ghost job window-feasible. At hour t , for each $k = 1, \dots, \ell$
 279 (the look-ahead), CARMA adds to the planning problem a *ghost job* for
 280 every (o, b) with work $\kappa \bar{W}_{\tau(t+k)ob}$, width $\kappa \bar{M}_{\tau(t+k)ob}$, release $t+k$ and deadline
 281 $\min(t+k+b, t+H)$. Ghost jobs compete with real jobs for capacity, but
 282 their slack arc is priced at a moderate penalty Π_g , set to 1.5 times the largest
 283 local unit cost in the window, rather than at Π . They can therefore reserve
 284 clean capacity when it is economical but never force real work to be late. The
 285 weight $\kappa \geq 0$ scales how much of the expected demand is anticipated. With

286 $\kappa = 0$, CARMA reduces to myopic MPC.

287 *Carbon cost view.* Future-hour costs in Eq. (1) are evaluated at a forecast of
 288 $I_s(t+h)$; for the current hour ($h = 0$) the observed intensity is used. CARMA
 289 admits two cost views: the point forecast $\hat{I}$, or the conformal upper quantile
 290 U^β of Section 3.2. The second is motivated by the optimizer’s curse [55]:
 291 selecting the minimum over noisy estimates is biased towards optimism. The
 292 conformal view replaces each future cost by a calibrated quantile, which
 293 shifts it by a site- and horizon-specific amount proportional to the local
 294 volatility. For $\beta > 0.5$ this penalises volatile sites and distant hours more than
 295 stable sites and near hours. Whether this risk premium helps is an empirical
 296 question, and we treat the cost view as a hyperparameter.

297 *Algorithm.* Fig. 1 summarises the loop. At each hour t , CARMA (i) observes
 298 $I_s(t)$, the free capacity and the pending jobs; (ii) builds the cost cube $c_{os}(t+h)$
 299 for $h < H$ from the chosen forecast view; (iii) adds the ghost jobs for hours
 300 $t+1, \dots, t+\ell$; (iv) solves the resulting min-cost flow problem exactly; (v)
 301 executes the first-hour allocations of the real jobs only; and (vi) after observing
 302 the realised intensities, updates the ACI levels by Eq. (7) when conformal
 303 costs are used. Overdue work gets a one-hour window and is served with
 304 priority. The planning network has $O((n_t + \ell|\mathcal{S}||\mathcal{B}|)H|\mathcal{S}|)$ arcs, where n_t is
 305 the number of pending jobs, so each decision takes a few tens of milliseconds
 306 (Section 5). The hyperparameters (κ, ℓ) and the cost view are tuned on
 307 validation weeks that precede the test period (Section 5.2).

308 3.4. Baselines

309 We compare CARMA with the following online policies and the oracle.
 310 **ASAP-Local** is carbon-agnostic: it runs work at its origin as soon as possible,
 311 earliest deadline first (EDF). **Greedy-Spatial** also runs work immediately,
 312 but at the site that is currently cleanest; this is geographical load balancing
 313 without temporal shifting [13]. **Forecast-Reserve** books, when a job arrives,
 314 the cheapest forecast (site, hour) slots in its window that still have capacity,
 315 and never revises the booking. This is the lowest-window strategy of [10]
 316 extended to multiple sites. **MPC** is the myopic receding-horizon flow scheduler
 317 with point forecasts. **MPC-Protect** adds a protection level in the spirit
 318 of revenue management: planned work may use only a share $1 - r$ of the
 319 capacity of the sites that are cleaner than the fleet median over the window.
 320 In the current hour, the protected share is first offered to jobs due within τ

hours and then, because unused capacity perishes, to all remaining work. The parameters (r, τ) are tuned on validation data. **MPC+conformal** uses the conformal upper quantile at the level that was best for MPC on validation data. **MPC-Perfect** is MPC given the realised intensities, which isolates the value of forecast accuracy. **CARMA-Perfect** is CARMA given the realised intensities, and **CARMA+conformal** is CARMA with the best conformal cost view. **Oracle** solves (2)–(5) with full knowledge of arrivals and intensities.

3.5. Data and experimental design

Carbon intensity. We use half-hourly regional carbon intensity for Great Britain from the Carbon Intensity API of the National Energy System Operator (NESO) [45] from 1 January 2022 to 31 December 2024. Values are averaged to hourly resolution. NESO does not meter regional intensity. For past periods, the regional endpoint returns a single modelled estimate per half-hour (in a field named `forecast`), and we treat this estimate as the realised intensity. The oracle and the “Perfect” variants are therefore clairvoyant with respect to NESO’s published regional values, and all emission figures inherit their modelling error. NESO’s forward regional forecasts are not archived, so they cannot serve as a historical benchmark for our forecaster. Four regions host the candidate sites: South Scotland, North West England, West Midlands and London. They span the north–south gradient of the British grid, from wind- and nuclear-rich South Scotland to gas-dependent London, and include the main data-centre markets of London, Manchester (North West England) and Birmingham (West Midlands). The largest cluster of the London market, around Slough, lies in NESO’s South England region; we use the London region as a stylised proxy for a southern, gas-dependent site. Gaps (602 of 26,304 hours in at least one region) were filled by linear interpolation. The period up to 30 June 2023 is used to train the forecaster, July–September 2023 to calibrate the conformal scores, October–December 2023 to tune the scheduler, and all 52 weeks starting on Mondays in 2024 as the out-of-sample test set.

Sites and energy parameters. Batch capacities are 300, 400, 400 and 600 servers for South Scotland, North West England, West Midlands and London. This reflects a small, clean northern site and a large site in London; the split is varied in the sensitivity analysis. Interactive load takes $20\% \pm 15\%$ of each site’s capacity on a diurnal cycle that peaks in the afternoon. We assume

357 $e = 0.4$ kWh per server-hour. The assumed PUE values are 1.15, 1.20, 1.25
 358 and 1.30 from north to south, reflecting free-cooling potential. Migrated work
 359 moves $\delta = 2$ GB per server-hour at $\eta = 0.02$ kWh/GB. This value lies within
 360 the wide range of published estimates of transmission energy intensity [62].
 361 The main experiments count only the energy of busy servers, which assumes
 362 that idle servers are switched off or reallocated. The sensitivity analysis varies
 363 η and the capacity split, and it adds an always-on case with an idle draw of
 364 0.2 kWh per server-hour for every batch server.

365 *Workloads.* Job arrivals follow a non-homogeneous Poisson process with a
 366 diurnal peak in the afternoon and 30% lower weekend rates. Jobs originate
 367 at the four sites with probabilities 0.10, 0.20, 0.25 and 0.45. Width m_j is
 368 log-uniform on $[4, 64]$ servers and the run time at full width r_j is uniform
 369 on $\{1, \dots, 6\}$ hours, so $w_j = m_j r_j$. A share v of jobs is urgent (deadline
 370 $= a_j + r_j$, no temporal slack); for the others the window is $\lceil f_j r_j \rceil$ hours with
 371 $f_j \sim U[1.5, 4]$, capped at 24 hours. The baseline value is $v = 0.3$. The arrival
 372 rate is scaled so that the offered load equals a fraction $\rho \in \{0.3, 0.5, 0.7\}$ of
 373 the mean free capacity. Each test episode covers one week of arrivals. We
 374 also simulate 24 hours of arrivals before and after the week, so the evaluated
 375 week starts and ends under steady-state congestion. Metrics are computed
 376 only for jobs released during the week. For each week and load level we
 377 draw three independent workload traces with fixed seeds, and every policy
 378 runs on the same traces. The demand profile used by CARMA is estimated
 379 from 13 independent historical traces of the same generator. This favours
 380 anticipation, because the profile is then an unbiased estimate of the arrival
 381 rate. We therefore also evaluate a trace-driven workload whose profile is
 382 learned from different days (next paragraph) and profiles learned at the wrong
 383 load (Section 5.5). We study two regimes: *spatio-temporal*, where migration
 384 is allowed, and *temporal-only*, where it is not. All hyperparameters are tuned
 385 separately for each regime.

386 *Trace-driven workload.* To test anticipation on real, non-stationary demand,
 387 we use the batch-task table of the Alibaba 2018 cluster trace [63, 64]. It
 388 records 3.85 million terminated batch jobs over eight days, with start and end
 389 times, numbers of instances and CPU requests. Each job’s work is its CPU
 390 core-hours and its width is its peak concurrent cores, both converted to servers
 391 of 96 cores. The work is then scaled so that the offered load in the evaluation
 392 days equals $\rho = 0.5$. The trace records no deadlines, so windows are drawn as

in the synthetic model: 30% of jobs are urgent, with a window equal to their run time rounded up to whole hours, and the others receive 1–23 hours of slack. Origins are assigned at random with the site probabilities above. Jobs are aggregated into batches by arrival hour, origin and window class, which leaves the fluid problem unchanged up to the pooling of widths. CARMA learns an hour-of-day profile from days 1–4 of the trace and is evaluated on days 5–8, preceded by 24 hours of warm-up. Offered work in days 5–8 is 39% higher than in days 1–4, and hourly work ranges from 0.2 to 3.6 times its mean, so the profile is biased and bursts are not anticipated. The four evaluation days are paired with each of the 52 test weeks of 2024, starting on the Monday, with three independent draws of origins and deadlines.

Metrics and statistics. The primary metric is operational emissions (tCO₂) per week under average regional intensity, which is an *attributional* measure (Section 6 discusses marginal emissions). We report the reduction relative to ASAP-Local, the optimality gap relative to the oracle, the share of work finished late, the share of work executed away from its origin and the decision time. Online policies keep processing late work after its deadline, and its emissions are counted. Because policies differ in lateness, we also report a service-adjusted gap, in which late work is additionally charged at the week’s largest local unit cost. Statistical tests use week-level means over the three traces, giving 52 pairs per comparison. Policies are compared by two-sided Wilcoxon signed-rank tests, with Holm correction applied jointly to all comparisons in a table. Consecutive weeks are serially correlated (lag-1 autocorrelation of the paired differences ≈ 0.5), so confidence intervals for mean paired differences use a moving-block bootstrap over weeks with blocks of four weeks [65].

4. Theoretical guarantees

This section establishes a worst-case limit for *every* online scheduler, prediction-dependent guarantees for CARMA, and a fairness property of the carbon attribution induced by the pooled scheduling problem. Proofs are in Appendix A. Throughout, unit costs (Eq. (1), including any backstop resource introduced below) satisfy $0 < L \leq c_{os}(t) \leq U$, and $\theta = U/L$ is the *carbon-intensity ratio*. Operational costs alone can be zero, because the hourly regional intensity of South Scotland is 0 gCO₂/kWh in 133 hours of our 2022–2024 data. A positive floor appears once the embodied emissions

of the servers are allocated per server-hour. As an illustration, a floor of 30 gCO₂e per server-hour and the largest operational unit cost observed in 2024 (221 g) give $\theta \approx 8.4$. An online algorithm is ρ -competitive if its emissions never exceed ρ times those of the clairvoyant oracle, for any arrival sequence and any intensity trajectory.

4.1. A universal lower bound under demand uncertainty

Competitive analyses of carbon-aware scheduling model uncertainty about future prices, that is, intensities, for a single workload. Examples are online search and one-way trading [24], the online pause-and-resume problem [25], and spatiotemporal allocation with deadlines (CarbonClipper) [26]. When several jobs share scarce low-carbon capacity, difficulty remains even when all prices are known in advance, because future demand is unknown.

Theorem 1 (Lower bound). *Let $\theta \geq 1$. No online algorithm (deterministic or randomised, with integral or fractional decisions) can be better than $R^*(\theta)$ -competitive, even if it knows all future carbon intensities exactly, where*

$$R^*(\theta) = \max_{1 \leq p \leq \theta} \left\{ 1 + \frac{(p-1)(\theta-p)}{p^2-p-1+\theta} \right\} \geq 1 + \frac{\sqrt{\theta}(\sqrt{\theta}-1)^2}{2\theta-\sqrt{\theta}-1} = \Omega(\sqrt{\theta}). \quad (8)$$

On the same family of instances, the competitive ratio of myopic receding-horizon scheduling (MPC) is at least $(1+\theta)/2$, a value approached as $p \rightarrow 1$.

The bound is established on a two-hour instance and is a quantitative version of Proposition 2, which treats the case $L = 0$. Any algorithm must decide how much flexible work to run now at moderate cost before it learns whether an urgent job will claim the cheap capacity later. The adversary thus raises the effective price of the clean hour from L to U . The resulting order, $\sqrt{\theta}$, is the same as for deterministic online min-search with unknown prices [24]. The difference is where it comes from: here every price is known, and the loss is caused only by the unknown arrival of a competing job. For the illustrative $\theta \approx 8.4$ above, $R^* = 1.80$, whereas myopic MPC can be driven to 4.7. For $\theta = 100$ the values are 5.29 and 50.5. Theorem 1 is a worst-case statement, but it points in the same direction as the average-case evidence of Section 5.3. Carbon forecasts cannot substitute for information about demand, and this motivates the learning-augmented design of CARMA [27].

4.2. Deadline safety, consistency and smoothness of CARMA

We analyse CARMA under two assumptions.

(A1) *Backstop capacity.* In every hour, work may also run on a backstop resource of unlimited capacity at a unit cost in $[L, U]$, for example by bursting to a public-cloud region or to reserve servers at a carbon-intensive site. All jobs, real and predicted, are *window-feasible*, $w_j \leq m_j(d_j - a_j)$. The penalties satisfy $\Pi > U$ for real jobs and $\Pi_g \geq U$ for ghost jobs. The implementation enforces window-feasible ghost jobs (Section 3.3) and sets Π_g to 1.5 times the largest local unit cost, which exceeds every unit cost including transfer overheads.

(A2) *Episode within the planning window.* The episode $\{0, \dots, T - 1\}$ has length $T \leq H$, as for a daily batch window, so every planning problem extends to the end of the episode.

To measure prediction quality at hour t , let F_t be the set of jobs that actually arrive after t , and $\hat{F}_t$ the ghost jobs used by CARMA. Jobs are compared after splitting them into *atoms* with proportional widths, which leaves every planning problem unchanged (Appendix A). Let d_t be the smallest total work of atoms that must be added to or removed from $\hat{F}_t$ to obtain F_t . Let $\varepsilon_t = \max_{\tau > t, o, s} |\hat{c}_{os}(\tau | t) - c_{os}(\tau)|$ be the carbon-cost forecast error, and W_t the total work pending at t plus the larger of the true and predicted future work.

Proposition 3 (Baseline robustness). *Under (A1), every policy that completes all work by its deadlines is θ -competitive.*

Theorem 2 (Guarantees for CARMA). *Under (A1), for any arrival sequence and any predictions, CARMA completes every job by its deadline (deadline safety) and is therefore θ -competitive. If (A2) also holds, then*

$$\text{ALG} \leq \min \left\{ \theta \cdot \text{OPT}, \text{OPT} + \sum_{t=0}^{T-1} [(U - L) d_t + 2 \varepsilon_t W_t] \right\}. \quad (9)$$

In particular, CARMA is optimal with exact predictions ($d_t = \varepsilon_t = 0$), which is consistency, and its excess emissions grow at most linearly in the prediction errors, which is smoothness.

These properties should be read with care. Robustness by itself does not distinguish CARMA: by Proposition 3 it holds for any policy that meets its deadlines. What CARMA adds is deadline safety under *arbitrary* ghost predictions, together with consistency and smoothness. Deadline safety holds because ghost jobs carry a softer penalty than real jobs, so a misprediction can divert real work to dirtier capacity but never make it late. The smoothness term charges demand errors at the spread $U - L$ of unit costs, not at the lateness penalty. The term is informative when predictions are close to the realised demand. For random arrivals, however, an expected-value profile never matches the realised jobs exactly. Moreover, each misprediction enters d_t in every epoch in which it is visible, so the bound in (9) can be loose. In the experiments, assumption (A2) is not satisfied (weekly episodes with $H = 24$ h), and no backstop is modelled. The theorem therefore describes the mechanism, not a certificate for the reported numbers; the degradation profile is tested empirically in Section 5.6. Robustness θ is also far from the $\Omega(\sqrt{\theta})$ lower bound. Closing this gap while retaining consistency is left for future work, for example by protecting deferral decisions with reservation prices [24, 26]. Fig. 2 summarises the worst-case landscape.

4.3. Fair carbon attribution among sites

Pooling capacity reduces total emissions but redistributes them. Work submitted in London is executed in South Scotland, and Scottish jobs may be displaced by it. The organisations that operate the sites, or the business units that submit the jobs, therefore need an attribution of the pooled emissions that each of them can accept. This is an *internal* allocation problem, like carbon transfer pricing between units. It is distinct from external GHG Protocol reporting, where Scope 2 emissions are reported by the entity that consumes the electricity [66].

We model the allocation as a cooperative game. The players are the sites $\mathcal{S}$. A coalition $C \subseteq \mathcal{S}$ pools the capacity of its sites and serves the jobs submitted at them. Its stand-alone value $c(C)$ is the optimum of (2)–(5) restricted to those jobs and sites, which includes the lateness penalty if the coalition cannot serve its own jobs on time. An attribution $\phi \in \mathbb{R}^{|\mathcal{S}|}$ is *budget-balanced* if $\sum_o \phi_o = c(\mathcal{S})$. It is in the *core* if $\sum_{o \in C} \phi_o \leq c(C)$ for every coalition C . A core attribution gives no group of sites a reason to leave the pool and schedule on its own.

Let (λ, ν, μ) be optimal dual variables of the grand-coalition problem: λ_j for the work constraint (3), $\nu_{jt} \leq 0$ for the width constraint (4) and $\mu_{st} \leq 0$

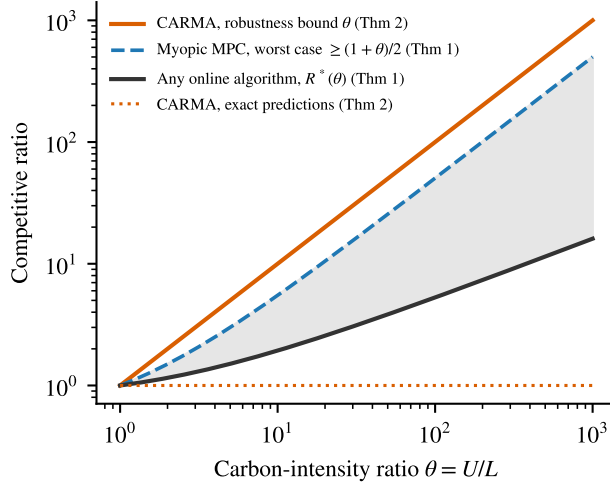


Figure 2: Worst-case landscape as a function of the carbon-intensity ratio θ . No online algorithm can lie below $R^*(\theta)$ (Theorem 1). Myopic MPC can be driven to $(1 + \theta)/2$ even with perfect carbon forecasts. Given a backstop, every policy that meets its deadlines, CARMA included, is within θ (Proposition 3). CARMA is optimal when its predictions are exact (Theorem 2).

524 for the capacity constraint (5). The *dual carbon attribution* is

$$\phi_o = \sum_{j: o_j=o} \left(\lambda_j w_j + \sum_t \nu_{jt} m_j \right) + \sum_t \mu_{ot} K_o(t). \quad (10)$$

525 Site o is charged the carbon shadow price of the work it submits and is
526 credited ($\mu \leq 0$) with the shadow value of the capacity it contributes. A
527 site whose clean capacity is worth more than the emissions of its own jobs
528 therefore receives a net credit, that is, a negative internal charge.

529 **Theorem 3** (Core-stable carbon attribution). *The attribution (10) is budget-*
530 *balanced and in the core of the pooling game. In particular, it is individually*
531 *rational: no site is charged more than its stand-alone optimum $c(\{o\})$.*

532 The result follows from Owen’s theorem on linear production games [29];
533 games defined by network flows are known to be totally balanced [30]. The
534 duals are those of the hindsight problem, which suits after-the-fact accounting.
535 They are available as the optimal node potentials of the min-cost flow; in
536 our experiments we obtain them from an LP solver. Optimal duals need not

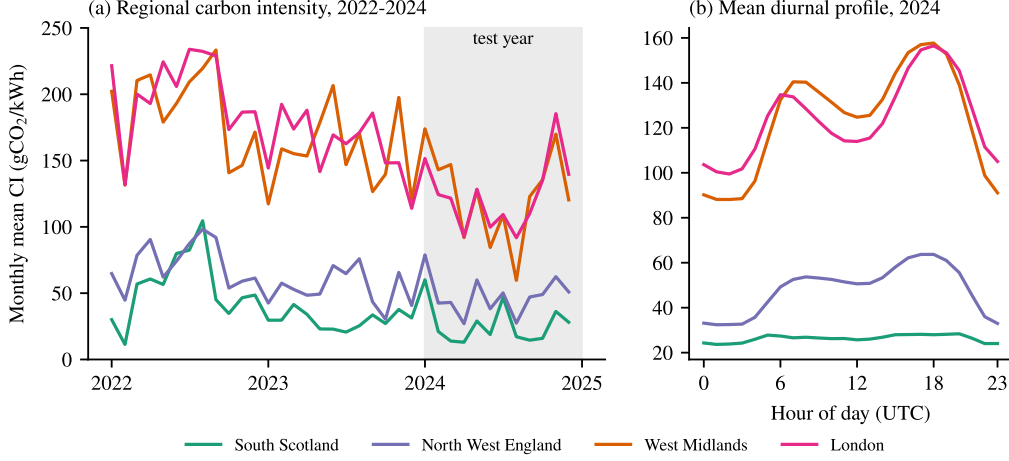


Figure 3: Hourly carbon intensity of the four candidate sites (NESO regional data). (a) Monthly means, 2022–2024; the shaded area is the out-of-sample test year. (b) Mean diurnal profile in 2024 (UTC).

be unique, so Section 5.7 reports the range of (10) over all optimal duals. It also compares the dual attribution with the Shapley value [67], which is budget-balanced but not guaranteed to lie in the core, and with simpler rules. The simpler rules are *origin-based* (each site is charged the emissions of its own jobs wherever they ran), *host-based* (each site is charged the emissions of the work it executes, the analogue of location-based Scope 2 reporting), and proportional rules.

5. Results

5.1. Carbon-intensity data and forecasts

Fig. 3 shows the regional intensities. Over 2022–2024, mean intensity was 37 gCO₂/kWh in South Scotland, 58 in North West England, 156 in the West Midlands and 163 in London. At any hour the gap between the cleanest and dirtiest site is therefore large, which is what makes spatial shifting attractive. Intensities fell steadily: London’s annual mean dropped from 202 gCO₂/kWh in 2022 to 124 in 2024. Diurnal profiles peak in the early evening (around 18:00 UTC), and the profile of South Scotland is much flatter than those of the southern sites. The four series are positively correlated (pairwise correlations between 0.50 and 0.87), because national wind output drives all of them.

555 The forecaster was trained on 1.24 million (issue time, site, horizon)
556 samples in about one minute. In 2024 its mean absolute error (MAE), averaged
557 over horizons and sites, is 31.2 gCO₂/kWh, against 34.4 for persistence and
558 40.4 for a seasonal-naive forecast. The advantage is about 23% at horizons
559 of 1–6 hours, and the difference is significant at all English sites up to 18
560 hours in Diebold–Mariano tests [68]. Day-ahead skill is limited, because
561 no weather inputs are used. Adaptive conformal inference keeps the upper
562 bounds calibrated despite the 2024 decarbonisation drift. Appendix B gives
563 the full evaluation (accuracy by horizon and site, coverage, pinball loss).

564 5.2. Hyperparameter tuning

565 All policies were tuned on 13 validation weeks (October–December 2023,
566 $\rho \in \{0.3, 0.5, 0.7\}$), separately for each regime. Fig. 4 shows the spatio-
567 temporal regime. Anticipating demand helps across the whole grid: every
568 tested configuration beats myopic MPC, whose mean gap to the oracle
569 is 22.3%. The gap is smallest around $\kappa = 1$, where expected demand is
570 anticipated without inflation: 9.0% for $\ell = 12$ and $\ell = 18$. Over-anticipation
571 ($\kappa = 1.25$ with $\ell = 18$) reserves too much clean capacity and is penalised
572 (12.1%). We selected $\kappa = 1$ and $\ell = 12$ hours, resolving configurations within
573 0.25 percentage points of the best towards the shorter look-ahead. In the
574 temporal-only regime, anticipation brings no benefit. The best configuration
575 ($\kappa = 0.5$, $\ell = 6$; 4.50%) is indistinguishable from MPC (4.50%), and larger
576 κ is slightly worse. For MPC-Protect, the best protection level was $r = 0.3$
577 with $\tau = 1$ h in the spatio-temporal regime (21.0% against 22.3% for MPC).
578 In the temporal-only regime every protection level was worse than MPC (at
579 best 6.0%).

580 The cost view interacts with anticipation (Fig. 4b). For myopic MPC,
581 high conformal levels help. The gap falls from 25.2% at $\beta = 0.5$ to 17.1%
582 at $\beta = 0.9$ and rises again at $\beta = 0.95$ (18.1%), against 22.3% with point
583 forecasts. For CARMA, every conformal level is worse than the point forecast:
584 the best, $\beta = 0.6$, gives 10.0% against 9.0%. The mechanism is not a uniform
585 risk premium. On the validation weeks, the $\beta = 0.6$ view lies on average
586 only 0.8 g below to 4.9 g above the point forecast at South Scotland, North
587 West England and London, but 17.8 g above it at the volatile West Midlands
588 site. The offset also grows with the horizon, and at $\beta = 0.8$ it rises to
589 28–128 g at a 24-hour horizon. High levels therefore make distant hours and
590 volatile sites look dirtier, so the planner runs work now and at stable sites,
591 which keeps future clean capacity free. For a myopic planner this is a crude

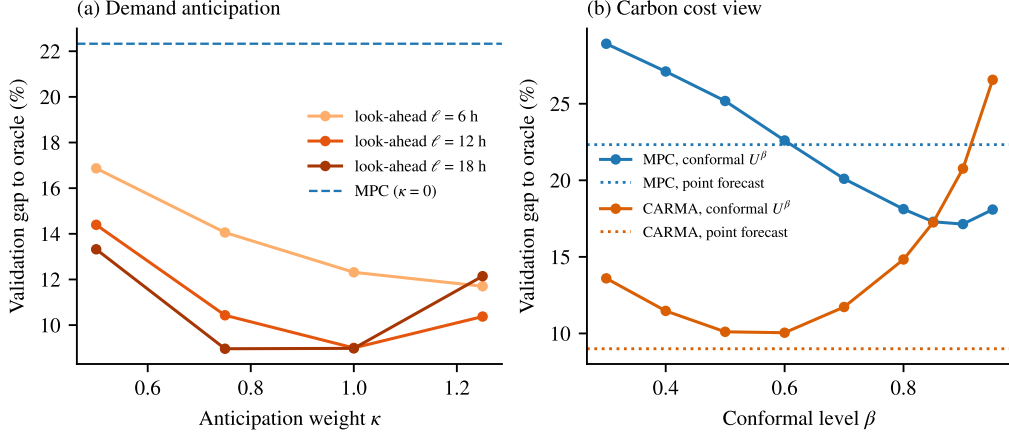


Figure 4: Validation results for the spatio-temporal regime (2023-Q4, mean over 13 weeks and three load levels). (a) Gap to the oracle as a function of the anticipation weight κ and look-ahead ℓ , using point forecasts. (b) Effect of the carbon cost view: conformal quantiles U^β against point forecasts, for myopic MPC and for CARMA ($\kappa = 1$, $\ell = 12$).

form of anticipation. Once CARMA represents future demand explicitly, the same distortion double-counts it and gives up useful deferrals. The selected CARMA therefore uses point forecasts. For the test we keep MPC+conformal at $\beta = 0.9$ and CARMA+conformal at $\beta = 0.6$ (spatio-temporal) as ablations. In the temporal-only regime, $\beta = 0.5$ was the best conformal view for both.

5.3. Out-of-sample scheduling performance

Tables 1 and 2 report results for the 52 test weeks of 2024, with three independent workload traces per week. Weekly traces contain on average 917, 1,527 and 2,141 jobs (69,400, 115,400 and 162,000 server-hours) at $\rho = 0.3$, 0.5 and 0.7. Fig. 5 shows the gaps to the oracle.

Spatio-temporal regime. With migration allowed, the oracle cuts attributional emissions relative to ASAP-Local by 70.8%, 57.5% and 40.6% at $\rho = 0.3$, 0.5 and 0.7. Greedy-Spatial realises only part of this potential (gaps of 38–66%), because it has no temporal flexibility and floods the cleanest site with whatever arrives first. The forecast-driven planners come closer. Forecast-Reserve has gaps of 24–27%, MPC 19–23%, and the two stronger baselines, MPC-Protect and MPC+conformal, 15–28%. MPC-Protect is the best baseline at $\rho = 0.5$ (16.1%) and MPC+conformal at the other loads (15.2% and 15.8%). CARMA

Table 1: Out-of-sample results in the spatio-temporal regime (migration allowed), 52 weeks of 2024×3 traces. Emis.: mean weekly operational emissions (tCO₂, attributional). Red.: reduction relative to ASAP-Local (%). Gap: excess over the clairvoyant oracle (%). Adj.: service-adjusted gap, in which late work is additionally charged at the week’s largest local unit cost. Late: share of work finished after its deadline (%). Mig.: share of work executed away from its origin (%). p : two-sided Wilcoxon signed-rank test of week-mean emissions against CARMA, Holm-adjusted over all comparisons in Tables 1 and 2 of the same regime. The best gap among implementable online policies is in bold.

Load	Policy	Emis.	Red.	Gap	Adj.	Late	Mig.	p
$\rho = 0.3$	ASAP-Local	3.70	0.0	280.1	280.1	0.00	0.0	<0.001
	Greedy-Spatial	1.75	54.1	65.5	65.5	0.00	81.7	<0.001
	Forecast-Reserve	1.38	63.9	26.3	26.3	0.00	83.6	<0.001
	MPC	1.31	65.8	19.2	19.2	0.00	69.4	<0.001
	MPC-Protect	1.40	63.4	28.0	28.0	0.00	80.5	<0.001
	MPC+conformal	1.27	66.8	15.2	15.2	0.00	70.0	<0.001
	MPC-Perfect	1.28	66.6	16.4	16.4	0.00	69.1	<0.001
	CARMA	1.22	68.3	9.3	9.3	0.00	70.1	–
	CARMA+conformal	1.22	68.3	9.5	9.5	0.00	70.4	0.037
	CARMA-Perfect	1.15	70.2	2.4	2.4	0.00	69.9	<0.001
	Oracle	1.13	70.8	0.0	0.0	0.00	70.9	–
$\rho = 0.5$	ASAP-Local	6.08	0.0	147.7	148.2	0.05	0.0	<0.001
	Greedy-Spatial	4.12	33.3	62.2	62.2	0.00	76.4	<0.001
	Forecast-Reserve	3.32	46.9	26.8	26.9	0.02	78.2	<0.001
	MPC	3.22	48.5	22.9	23.0	0.01	50.0	<0.001
	MPC-Protect	3.07	51.1	16.1	16.1	0.00	58.7	<0.001
	MPC+conformal	3.08	51.1	16.4	16.4	0.00	50.6	<0.001
	MPC-Perfect	3.18	49.3	21.1	21.2	0.01	49.4	<0.001
	CARMA	2.92	53.6	9.8	9.8	0.00	50.5	–
	CARMA+conformal	2.94	53.4	10.3	10.3	0.00	50.6	<0.001
	CARMA-Perfect	2.79	56.0	3.8	3.8	0.00	50.2	<0.001
	Oracle	2.70	57.5	0.0	0.0	0.00	51.9	–
$\rho = 0.7$	ASAP-Local	8.29	0.0	71.7	98.5	4.54	0.0	<0.001
	Greedy-Spatial	6.76	19.1	37.9	37.9	0.00	73.3	<0.001
	Forecast-Reserve	6.14	26.9	24.0	30.6	1.12	73.5	<0.001
	MPC	5.94	29.3	20.0	30.9	1.89	31.6	<0.001
	MPC-Protect	5.82	30.9	17.1	21.0	0.69	37.8	<0.001
	MPC+conformal	5.76	31.7	15.8	17.9	0.34	32.1	<0.001
	MPC-Perfect	5.87	30.1	18.6	30.1	1.93	31.4	<0.001
	CARMA	5.41	36.1	8.0	8.1	0.02	31.5	–
	CARMA+conformal	5.44	35.8	8.4	8.5	0.01	31.6	<0.001
	CARMA-Perfect	5.24	38.4	3.9	4.2	0.04	31.0	<0.001
	Oracle	5.07	40.6	0.0	0.0	0.00	32.8	–

Table 2: Out-of-sample results in the temporal-only regime (no migration), 52 weeks of 2024 $\times$ 3 traces. Columns as in Table 1. Greedy-Spatial is omitted because without migration it coincides with ASAP-Local.

Load	Policy	Emis.	Red.	Gap	Adj.	Late	p
$\rho = 0.3$	ASAP-Local	3.70	0.0	21.8	21.8	0.00	<0.001
	Forecast-Reserve	3.27	11.8	7.1	8.6	0.38	<0.001
	MPC	3.23	13.0	5.6	5.6	0.00	0.743
	MPC-Protect	3.30	11.3	7.8	7.8	0.00	<0.001
	MPC+conformal	3.24	12.9	5.7	5.8	0.01	0.232
	MPC-Perfect	3.09	17.4	0.2	0.3	0.01	<0.001
	CARMA	3.23	13.0	5.6	5.6	0.00	–
	CARMA+conformal	3.24	13.0	5.7	5.7	0.01	0.232
	CARMA-Perfect	3.09	17.4	0.2	0.2	0.01	<0.001
	Oracle	3.08	17.6	0.0	0.0	0.00	–
$\rho = 0.5$	ASAP-Local	6.08	0.0	17.6	17.8	0.05	<0.001
	Forecast-Reserve	5.54	9.2	6.6	18.3	3.06	<0.001
	MPC	5.47	10.5	5.1	6.9	0.53	<0.001
	MPC-Protect	5.57	8.8	7.1	8.8	0.50	<0.001
	MPC+conformal	5.50	9.9	5.7	8.4	0.75	<0.001
	MPC-Perfect	5.30	13.7	1.3	4.1	0.76	<0.001
	CARMA	5.46	10.7	4.8	6.1	0.38	–
	CARMA+conformal	5.48	10.2	5.3	7.3	0.54	<0.001
	CARMA-Perfect	5.28	14.0	0.9	2.9	0.54	<0.001
	Oracle	5.24	14.8	0.0	0.0	0.01	–
$\rho = 0.7$	ASAP-Local	8.29	0.0	9.4	22.3	4.54	<0.001
	Forecast-Reserve	7.94	4.3	4.7	30.7	8.15	<0.001
	MPC	7.88	5.0	3.9	42.6	11.95	0.001
	MPC-Protect	8.00	3.6	5.4	43.0	11.62	<0.001
	MPC+conformal	7.92	4.5	4.5	49.6	13.53	<0.001
	MPC-Perfect	7.77	6.5	2.3	49.4	13.83	<0.001
	CARMA	7.87	5.1	3.7	32.6	9.20	–
	CARMA+conformal	7.90	4.7	4.2	39.8	10.87	<0.001
	CARMA-Perfect	7.75	6.8	1.9	41.2	11.63	<0.001
	Oracle	7.62	8.5	0.0	0.0	0.75	–

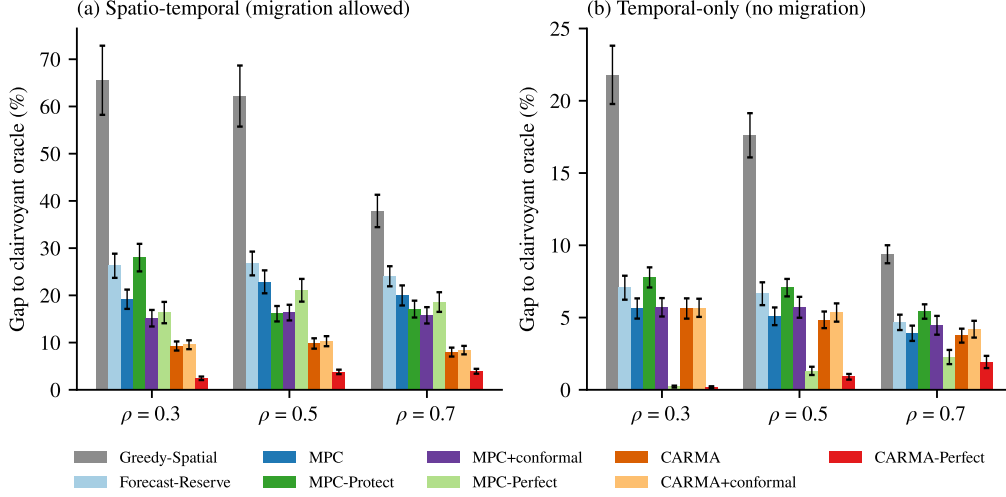


Figure 5: Mean gap to the clairvoyant oracle over the 52 test weeks of 2024 (week means of three traces) with 95% confidence intervals. ASAP-Local is omitted (gap 72–280% with migration, 9–22% without).

reduces emissions by 68.3%, 53.6% and 36.1% and has gaps of 9.3%, 9.8% and 8.0%. Relative to MPC this is a further emission reduction of 7.4%, 9.3% and 8.8%. Its advantage over the best baseline at each load is 5.9 (95% block-bootstrap CI 4.3–8.2), 6.3 (5.2–7.3) and 7.8 (6.7–8.9) percentage points of gap. CARMA emits less than MPC in 51, 51 and 52 of the 52 week-means. All comparisons with online baselines are significant after Holm correction ($p < 0.001$). CARMA+conformal is 0.3–0.5 points worse than CARMA ($p \leq 0.04$).

CARMA also finishes more work on time under heavy load. At $\rho = 0.7$, 0.03% of its work is late, against 1.9% for MPC, 0.7% for MPC-Protect, 1.1% for Forecast-Reserve and 4.5% for ASAP-Local. The service-adjusted gap therefore widens the difference: 8.1% for CARMA against 17.9–30.9% for the forecast-driven baselines. CARMA migrates about the same share of work as MPC (31–70%), whereas MPC-Protect migrates more (38–81%). CARMA’s advantage thus comes from better timing of the use of scarce clean capacity, not from moving more data. For the fleet of 1,700 batch servers at $\rho = 0.5$, CARMA emits 0.30 tCO₂ per week less than MPC, about 16 tCO₂ per year, and 3.16 tCO₂ per week less than carbon-agnostic operation. These absolute figures scale with fleet size.

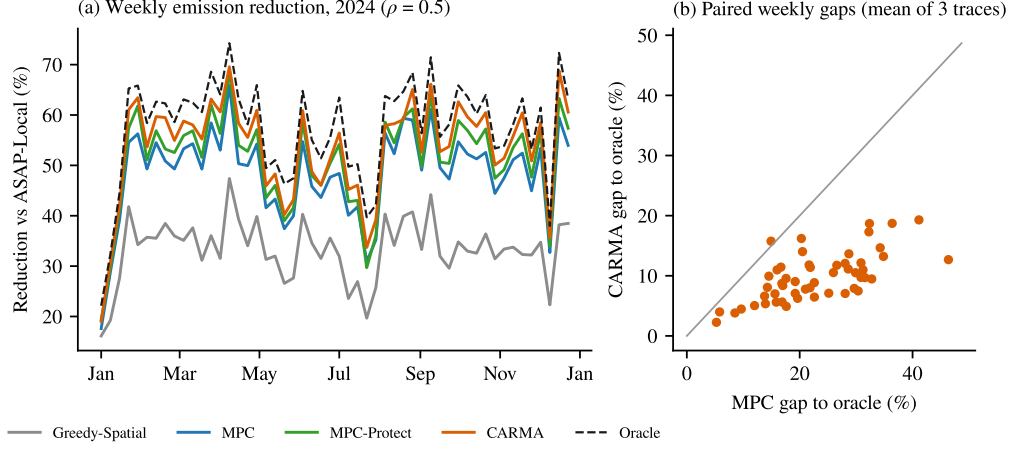


Figure 6: Weekly results in 2024 (spatio-temporal regime, $\rho = 0.5$, means of three traces). (a) Emission reduction relative to ASAP-Local. (b) Paired weekly gaps to the oracle; points below the diagonal are weeks in which CARMA outperforms MPC.

629 *Value of information.* The clairvoyant variants separate two sources of loss.
630 With perfect intensity forecasts, the myopic MPC-Perfect still has gaps of
631 16.4%, 21.1% and 18.6%. Perfect carbon foresight is thus worth only 1.4–
632 2.8 percentage points to a myopic planner, whereas anticipating demand is
633 worth 9.9–13.1 points (MPC versus CARMA). With both, CARMA-Perfect
634 comes within 2.4–3.9% of the oracle. The two kinds of information are
635 complementary. Once demand is anticipated, forecast accuracy is worth 4.1–
636 6.8 points (CARMA versus CARMA-Perfect), about two to four times more
637 than for the myopic planner. The ranking is stable across seasons (Fig. 6a),
638 although the level of achievable reductions varies from week to week with
639 wind conditions.

640 *Temporal-only regime.* Without migration (Table 2), each job can only move
641 in time at its origin. Achievable reductions are much smaller: the oracle saves
642 17.6%, 14.8% and 8.5%. MPC and CARMA are within 3.7–5.6% of the oracle.
643 They are statistically indistinguishable at $\rho = 0.3$ (difference 0.00 points, CI
644 -0.02 to 0.02). CARMA is slightly better at higher loads, by 0.24 (0.15–0.37)
645 and 0.16 (0.08–0.27) points. Both are better than Forecast-Reserve and MPC-
646 Protect. Anticipation brings little benefit here, because each site’s capacity
647 serves only its own jobs, and flexible and urgent work rarely compete for the
648 same clean hours in a way that deferral cannot absorb. The remaining loss is

649 due almost entirely to forecast error: the Perfect variants close the gap to
650 0.2–2.3%. Heavy load exposes a trade-off that the emission metric hides. At
651 $\rho = 0.7$, the capacity at the origin sites is itself insufficient, and every carbon-
652 aware policy delays work: 9.2% of CARMA’s work and 12.0% of MPC’s
653 is late, against 4.5% for ASAP-Local. Under the service-adjusted metric,
654 carbon-agnostic operation (22.3%) is then preferable to MPC (42.6%) and
655 CARMA (32.6%). Without migration and near saturation, temporal shifting
656 should therefore be restricted or combined with explicit service constraints.

657 *Computation.* A CARMA decision takes 9–39 ms on one CPU core, against
658 2–6 ms for MPC; the difference comes from the ghost jobs, which enlarge
659 the planning network. This is negligible for hourly decisions. A clairvoyant
660 weekly oracle with up to about 2,800 jobs is solved exactly in under one
661 second by network simplex.

662 5.4. Trace-driven workload

663 Table 3 reports the Alibaba-trace experiment. Arrivals and job sizes are
664 taken from the trace, and CARMA’s profile is learned from different days,
665 so the profile is biased by 39% and cannot anticipate bursts. The evaluation
666 covers four trace days paired with 52 carbon windows of 2024, with three draws
667 of origins and deadlines each, and 3,072 job batches (65,300 server-hours) per
668 window. Real, bursty demand makes all policies worse than in the synthetic
669 experiment, but the ranking is unchanged. CARMA has a gap of 20.7%.
670 MPC has 36.1%, and MPC-Protect, the best baseline here, 26.7%. Carbon
671 foresight again matters less than demand anticipation: MPC-Perfect reaches
672 only 33.3%, and CARMA’s advantage over MPC-Protect is 6.0 percentage
673 points (95% CI 4.4–8.1; CARMA better in 42 of 52 weeks, $p < 0.001$). The
674 bias of the learned profile costs 2.7 points (CI 1.1–3.8): CARMA with a
675 profile estimated in-sample from the evaluation days reaches 18.0%. With the
676 biased profile but perfect carbon forecasts, the gap would fall by 3.7 points
677 (CARMA-Perfect, 17.0%). The remaining distance to the oracle reflects bursts
678 that no profile anticipates. CARMA also keeps late work low (0.10%, against
679 0.65% for MPC).

680 5.5. Sensitivity analysis

681 Table 4 varies one factor at a time around the base case: spatio-temporal
682 regime, $\rho = 0.5$, $v = 0.3$, $\eta = 0.02$ kWh/GB, capacities 300/400/400/600.
683 Each setting uses 26 alternating test weeks of 2024 with two fresh workload
684 traces per week. All policies keep their base-case hyperparameters.

Table 3: Trace-driven workload (Alibaba 2018 cluster trace, days 5–8; profile learned from days 1–4), paired with 52 carbon windows of 2024×3 draws of origins and deadlines; spatio-temporal regime, $\rho = 0.5$. Emissions are per four-day window. Columns as in Table 1; the service-adjusted gap is omitted.

Policy	Emis.	Red.	Gap	Late	Mig.	p
ASAP-Local	3.66	0.0	195.3	0.00	0.0	<0.001
Greedy-Spatial	2.50	32.9	92.5	0.00	75.0	<0.001
Forecast-Reserve	1.94	48.6	42.5	0.81	77.4	<0.001
MPC	1.86	50.9	36.1	0.65	52.8	<0.001
MPC-Protect	1.76	53.8	26.7	0.07	64.0	<0.001
MPC-Perfect	1.82	51.9	33.3	0.63	52.3	<0.001
CARMA	1.69	55.7	20.7	0.10	53.2	–
CARMA-Perfect	1.64	57.2	17.0	0.15	52.2	<0.001
Oracle	1.44	62.8	0.0	0.00	56.7	–
CARMA (in-sample profile)	1.65	56.6	18.0	0.01	52.2	<0.001

685 *Workload and network.* CARMA has the smallest gap of all online policies in
686 every setting. Except under marginal accounting (discussed below), it emits
687 less than MPC in all 26 weeks of each setting. The benefit of anticipation
688 grows with the share of urgent work, because urgent arrivals are precisely the
689 demand that myopic planners crowd out. With $v = 0.5$, CARMA’s gap is
690 6.7% against 14.9% for MPC-Protect and 21.5% for MPC. With $v = 0.1$, it is
691 11.9% against 16.0% and 23.0%. The ranking is unchanged over a twelve-fold
692 range of transmission energy intensity ($\eta = 0.005$ – 0.06 kWh/GB), and the
693 migrated share of CARMA’s work stays at 49–51%.

694 *Capacity layout.* The advantage does not depend on the chosen layout. When
695 the clean South Scotland site is halved (150 servers) or doubled (600), or
696 London’s capacity is doubled, CARMA’s gap stays at 9.2–9.5%. The gaps
697 of MPC (21–28%) and MPC-Protect (16.6–18.5%) stay about two to three
698 times larger. More clean capacity raises the oracle’s reduction (from 51.6%
699 to 67.7%), but a myopic planner still misallocates it.

700 *Misspecified demand profile.* CARMA’s ghost jobs come from a profile learned
701 at $\rho = 0.5$. When the actual load is 20% higher ($\rho = 0.6$), so that CARMA
702 under-anticipates, its gap is 8.7% against 18.7% for MPC-Protect and 23.8%
703 for MPC. When the actual load is 20% lower ($\rho = 0.4$), CARMA over-
704 anticipates and reserves more clean capacity than needed. Its gap rises to
705 12.1%, but it still beats MPC-Protect by 3.2 points (15.3%) and MPC by

Table 4: Sensitivity analysis (spatio-temporal regime; 26 alternating weeks of 2024×2 traces per setting). Oracle red.: emission reduction of the oracle relative to ASAP-Local (%). Gap columns: excess over the oracle (%). Wins: weeks in which CARMA emits less than MPC. Late: share of late work (%), MPC / CARMA. ^aBusy servers draw 0.2 kWh above an idle draw of 0.2 kWh per server-hour that every batch server incurs regardless of the schedule. ^bSchedules are made with average intensities, but emissions are accounted with the stylised marginal emission factor of Section 5.5, and the oracle optimises marginal emissions.

Setting	Oracle	Gap to oracle (%)					Wins vs		Late (%)	
	red.	Greedy-Sp.	F.-Reserve	MPC	MPC-Prot.	CARMA	MPC	MPC / CARMA	MPC / CARMA	
Urgent share $v = 0.1$	60.8	77.5	28.3	23.0	16.0	11.9	26/26		0.00 / 0.00	
Urgent share $v = 0.5$	55.4	52.7	25.4	21.5	14.9	6.7	26/26		0.09 / 0.00	
Network energy $\eta = 0.005$ kWh/GB	61.2	68.7	28.5	27.0	17.5	10.9	26/26		0.01 / 0.00	
Network energy $\eta = 0.06$ kWh/GB	52.7	64.1	28.8	19.3	14.5	8.4	26/26		0.01 / 0.00	
Load $\rho = 0.4$, profile learned at 0.5	66.0	74.4	28.9	23.2	15.3	12.1	26/26		0.00 / 0.00	
Load $\rho = 0.6$, profile learned at 0.5	50.3	54.8	27.5	23.8	18.7	8.7	26/26		0.31 / 0.01	
Clean-site capacity halved (150)	51.6	57.1	25.2	21.4	16.6	9.5	26/26		0.02 / 0.00	
Clean-site capacity doubled (600)	67.7	83.0	32.4	28.4	16.9	9.2	26/26		0.01 / 0.00	
London capacity doubled (1200)	46.7	52.0	27.5	22.3	18.5	9.5	26/26		0.00 / 0.00	
Always-on servers (idle 0.2 kWh) ^a	18.2	10.7	4.7	3.5	2.5	1.5	26/26		0.01 / 0.00	
Marginal-emission accounting ^b	13.5	12.0	10.7	8.5	9.1	7.9	17/26		0.01 / 0.00	

11.1 points. The trace experiment of Section 5.4 adds a real, larger (39%) bias. The planner never accepts lateness of real work in exchange for serving ghost demand, so misprediction affects emissions, not deadlines.

Always-on servers. If every batch server draws idle power whether or not it is busy, a large part of the emissions no longer depends on the schedule. The achievable reduction falls to 18.2% for the oracle and 16.9% for CARMA. The ranking is unchanged (gaps of 1.5% for CARMA, 2.5% for MPC-Protect and 3.5% for MPC). Carbon-aware scheduling is therefore most valuable when it is combined with right-sizing, so that deferred work actually reduces the number of powered servers [18].

Marginal emissions. All sites are in one GB market, where gas-fired plant is usually the marginal generator [7, 46]. Moving load between regions then mainly changes *which* region is assigned the average mix, and it abates emissions physically only when low-carbon output would otherwise be curtailed. We therefore account emissions with a stylised marginal emission factor (MEF). The MEF is 394 gCO₂/kWh (combined-cycle gas) in every hour except two cases where it is zero. For South Scotland it is zero when Scotland is nearly carbon-free (below 25 g) while GB-wide intensity exceeds 100 g, a proxy for export-constrained hours; this holds in 27% of hours. For all sites it is zero when GB-wide intensity is below 50 g (4% of hours). Under this accounting

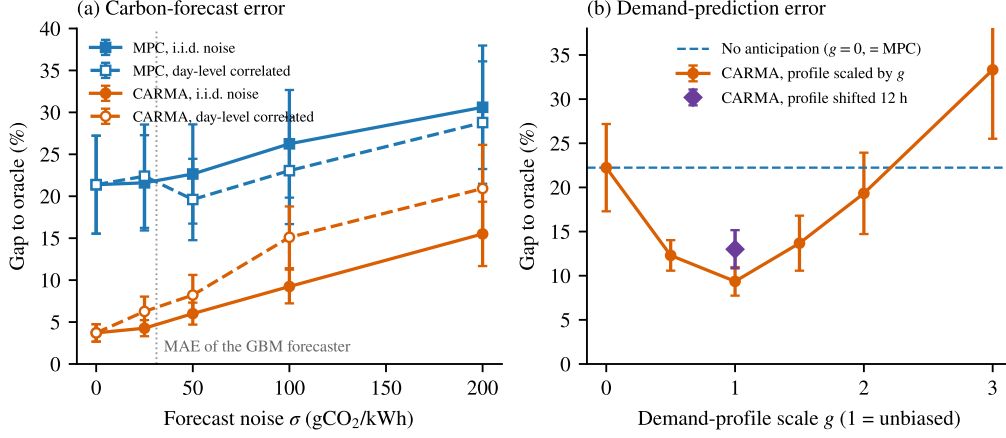


Figure 7: Degradation under controlled prediction errors (13 test weeks of 2024, $\rho = 0.5$, migration allowed; means with 95% confidence intervals). (a) Carbon-intensity forecasts replaced by the realised values plus Gaussian noise of standard deviation σ . The noise is either independent across issue times and horizons (solid) or shared by all forecasts of the same site and target day (dashed, open markers). (b) Demand profile scaled by g , or shifted by 12 hours so that arrivals are predicted at the wrong time of day.

the picture changes substantially. The oracle cuts marginal emissions by only 13.5%, and CARMA by 7.3%. The forecast-driven policies are within 7.9–10.7% of the marginal oracle. CARMA is still the best of them, but its advantage over MPC shrinks to 0.6 points (block-bootstrap CI 0.06–0.91), which is not significant after Holm correction ($p = 0.12$; CARMA better in 17 of 26 weeks). The schedules were optimised for average intensity, not for the MEF. The result nonetheless shows that most of the attributional reductions reported above are reallocations within the GB system. The physical benefit depends on whether shifting absorbs otherwise-curtailed low-carbon output, and a scheduler should be driven by a marginal signal when that is the goal.

5.6. Robustness to prediction errors

We test how CARMA degrades as its predictions deteriorate on every fourth test week of 2024 (13 weeks, $\rho = 0.5$, migration allowed, fresh workload traces). Controlled errors are injected into the two kinds of prediction (Fig. 7). These experiments do not satisfy the assumptions of the bound in Theorem 2, so they probe the degradation profile empirically.

742 *Carbon-forecast error.* With exact forecasts, CARMA’s gap is 3.7% and
743 MPC’s 21.4%. Independent noise is comparatively benign: CARMA’s gap
744 rises smoothly to 4.3%, 6.0%, 9.2% and 15.5% at $\sigma = 25, 50, 100$ and
745 200 gCO₂/kWh. Noise that persists over a whole target day, as when a
746 weather front is missed, is more damaging. The gaps are then 6.3%, 8.2%,
747 15.1% and 20.9%, because the planner cannot average out the error by re-
748 planning. MPC deteriorates from 21.4% to 28.8–30.6% at the largest noise
749 level. At every noise level and type, CARMA remains better than MPC with
750 the same forecasts. Only with the most persistent large errors ($\sigma = 200$) does
751 it fall to the level of MPC with *perfect* forecasts. The actual GBM forecaster
752 (MAE 31 gCO₂/kWh, with errors that are both serially and cross-sectionally
753 correlated) gives a gap of 9.8% in the main experiment. That lies between the
754 independent and correlated cases at comparable error size, so the degradation
755 should not be read as merely i.i.d.

756 *Demand-prediction error.* Scaling the learned demand profile by g produces a
757 V-shaped response with its minimum at the unbiased profile ($g = 1$, gap 9.4%).
758 Under-prediction by half ($g = 0.5$) and over-prediction by half ($g = 1.5$) cost
759 2.9 and 4.3 percentage points. Predicting arrivals at the wrong time of day
760 (a 12-hour shift) costs 3.6 points. Removing anticipation ($g = 0$) returns
761 the MPC level (22.2%). Gross over-prediction ($g = 3$, three times the true
762 demand) is worse than no anticipation (33.3%), because CARMA then holds
763 clean capacity for work that never arrives. A 12-hour shift keeps most of the
764 benefit, which indicates that much of the value comes from anticipating the
765 *volume* of demand rather than its exact timing. Late work was negligible
766 in all 325 runs (at most 0.01% of work). The largest ratio of CARMA’s
767 emissions to the oracle’s over all runs was 1.58.

768 5.7. Fair carbon attribution

769 We evaluate Theorem 3 on the 52 test weeks at $\rho = 0.5$, with the four
770 sites as players. For every week and each of the 15 coalitions we compute
771 the hindsight value $c(C)$, and we also run CARMA with only the coalition’s
772 sites and jobs. Pooling pays: the hindsight optimum of the grand coalition
773 emits 3.37 tCO₂ per week, against 6.70 for the four sites operating separately.
774 With CARMA the figures are 3.63 and 6.88. The question is how to share
775 the reduction.

776 We compare six attributions. The *origin-based* rule charges each site
777 the emissions of its own jobs wherever they run. The *host-based* rule, the

Table 5: Coalitional stability of carbon attributions (52 test weeks of 2024, $\rho = 0.5$). Coalition checks: share of the 52×14 (week, proper coalition) pairs in which the coalition is charged more than its stand-alone value. Weeks: share of weeks with at least one violation. IR: share of (week, site) pairs in which a site is charged more than its stand-alone value. Neg.: share of (week, site) pairs with a net credit. Max. excess: largest violation (tCO₂).

Benchmark	Attribution	Coalition checks (%)	Weeks (%)	IR (%)	Neg. (%)	Max. excess
Hindsight optimum	Origin-based	46.4	100.0	49.5	0.0	1.25
	Host-based	49.2	100.0	50.0	0.0	4.99
	Work-proportional	44.4	100.0	45.7	0.0	1.43
	Stand-alone proportional	22.0	100.0	0.0	0.0	0.84
	Shapley	3.7	44.2	0.0	40.9	0.20
	Dual (Theorem 3)	0.0	0.0	0.0	37.5	–
Hindsight, emissions only	Shapley	3.3	42.3	0.0	40.9	0.10
	Dual (Theorem 3)	0.0	0.0	0.0	37.5	–
CARMA (realised)	Origin-based	43.1	100.0	44.2	0.0	1.20
	Host-based	47.5	100.0	50.0	0.0	4.78
	Work-proportional	43.3	100.0	46.2	0.0	1.48
	Stand-alone proportional	21.3	100.0	0.0	0.0	0.86
	Shapley	3.8	50.0	0.0	39.9	0.13
	Dual (Theorem 3)	0.8	11.5	0.0	34.6	0.06

778 analogue of location-based Scope 2 accounting, charges each site the emissions
779 of the work it hosts. The *work-proportional* rule splits the total in proportion
780 to submitted work, and the *stand-alone proportional* rule in proportion to
781 stand-alone emissions. The *Shapley value* [67] averages marginal contributions
782 over all orders of joining. The *dual* rule is Eq. (10). Because $c(C)$ includes
783 the lateness penalty whenever a coalition cannot serve all of its work, we
784 also evaluate an emissions-only game, in which $c(C)$ is the emissions of the
785 hindsight schedule. Stand-alone coalitions leave work unserved in 94% of
786 weeks, although the amount is small. In that game the dual attribution is
787 rescaled to the emissions total and is no longer covered by the theorem. For
788 CARMA’s realised emissions, the dual attribution is applied to the hindsight
789 value and the online excess over the hindsight optimum is split in proportion
790 to work. A check is violated when a coalition is charged more than it would
791 emit on its own.

792 Table 5 and Fig. 8 report the results. In the hindsight game the dual
793 attribution satisfies all 728 coalition checks, as Theorem 3 guarantees. The
794 dual prices are not always unique, but over all optimal duals (checked in 9
795 weeks) the attribution of a site varies by at most 0.011 tCO₂ per week, so
796 the choice of dual does not matter in practice. The four accounting rules
797 violate 22–49% of the checks and fail in every week. The origin-based rule is
798 individually irrational for South Scotland in all 52 weeks and for North West
799 England in 51: their own jobs are displaced to dirtier capacity by imported

work, so they are charged more than they would emit on their own. The host-based rule fails for the same two sites in every week, because they are charged for the imported work they host; North West England, for example, is charged 1.50 tCO₂ per week against 0.59 on its own. Scaling stand-alone emissions down is individually rational by construction, but it still leaves 22% of the coalition checks violated. The Shapley value comes close: it is individually rational and violates 3.7% of the checks (44% of weeks), with a largest excess of 0.20 tCO₂. Its site charges are similar to the dual ones. Both credit the clean sites for the capacity they contribute. Under the dual rule South Scotland receives a net credit of 0.91 tCO₂ per week (Shapley 0.77), North West England is close to neutral, and London’s charge (2.81 t) remains well below its stand-alone emissions (3.91 t). Net credits arise in 38% of the (week, site) pairs. The Shapley value also needs all 2ⁿ coalition values, whereas the dual rule comes from a single solve. The emissions-only game gives almost the same picture: the rescaled dual attribution passes every check, and Shapley fails 3.3%.

With CARMA’s realised emissions, which is the practical case, the adjusted dual attribution remains individually rational for every site in every week. It violates 6 of the 728 coalition checks (0.8%, in 6 of 52 weeks), all involving North West England and London, joined in five of them by South Scotland, and the largest excess is 0.06 tCO₂. Shapley applied to the online values fails 3.8% of the checks, and the four accounting rules fail 21–48%. Stand-alone coalitions under CARMA leave at most 4.7% of their work late, and unfinished work emits nothing, so the online checks, if anything, favour secession.

6. Discussion

6.1. Managerial implications

Protect scarce clean capacity. In multi-site carbon-aware scheduling, the largest operational loss does not come from imperfect carbon forecasts. It comes from spending low-carbon capacity on work that could have waited or run elsewhere. Proposition 2 and Theorem 1 show this in the worst case, and the experiments confirm it on average. With migration allowed, a myopic planner with perfect carbon foresight is still 16–21% above the optimum in the synthetic experiment and 33% above it on the Alibaba trace. Clean capacity at a low-carbon site should be treated as a scarce resource with an opportunity cost, as in revenue management. A simple protection level (MPC-Protect) captures part of this value. CARMA captures more, because

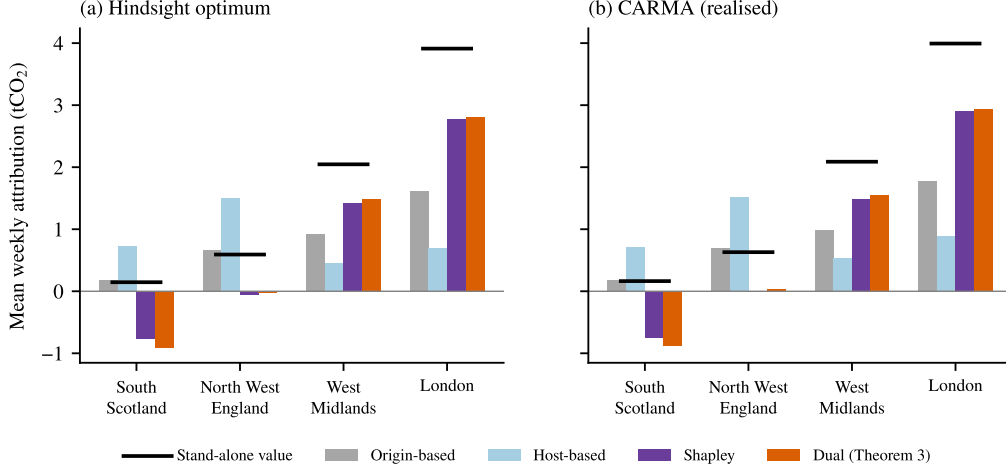


Figure 8: Mean weekly carbon attribution by site under four rules, compared with each site’s stand-alone value (black lines), 52 test weeks of 2024 at $\rho = 0.5$. (a) Hindsight optimum. (b) CARMA’s realised emissions. Negative values are net credits for contributing low-carbon capacity.

its ghost jobs price the protection hour by hour from the operator’s own history. The result held when the profile was biased by 39% and demand was bursty.

Invest in forecasts where they pay. The value of better intensity forecasts depends on the rest of the policy. For a myopic multi-site planner, perfect forecasts are worth only 1–3 percentage points; once demand is anticipated they are worth 4–7 points; without migration they are worth almost the whole remaining gap. Persistent errors, such as a missed day-ahead wind front, are more costly than independent ones. Forecasting effort, for example the integration of numerical weather predictions [44, 43], should therefore target day-level biases, and in multi-site settings it should follow the fix of demand myopia.

Use risk premia with care. The conformal cost views help a myopic planner but not CARMA. They work by making volatile sites and distant hours look dirtier, which crudely imitates anticipation. Once demand is anticipated, the same distortion gives up useful deferrals. Calibrated quantiles are better suited to reporting and to service-level decisions than to reshaping costs by default.

854 *Emissions or service near saturation.* Without migration and at high load,
855 every carbon-aware policy delayed work, and under a service-adjusted metric
856 carbon-agnostic operation was preferable. Operators should combine temporal
857 shifting with explicit service constraints, or restrict it, when local capacity is
858 close to saturation.

859 *Attributional versus physical savings.* The headline reductions are attribu-
860 tional: they measure how much of the work runs where and when the *average*
861 regional mix is clean. Within a single market such as Great Britain, moving
862 load between regions mostly reassigns the average mix. Under a stylised
863 marginal accounting, the achievable reduction fell from 58% to 14% and
864 CARMA’s advantage over MPC became small. Carbon-aware shifting is phys-
865 ically most valuable when it absorbs low-carbon output that would otherwise
866 be curtailed, for example behind transmission constraints between Scotland
867 and England, or when sites lie in different markets. In such settings a marginal
868 signal should drive the scheduler. The scheduling framework is unchanged:
869 only the cost coefficients of the flow network change. Idle power works in the
870 same direction. If servers stay powered, deferral saves only their dynamic
871 energy, and carbon-aware scheduling should be paired with right-sizing [18].

872 *Fair internal carbon accounting.* Pooling halved the hindsight emissions of
873 the four sites. However, origin-based accounting, which charges each site for
874 its own jobs wherever they run, would have made South Scotland worse off
875 than on its own in every week. A pool of organisations or business units
876 allocated this way is unstable, because its cleanest members have an incentive
877 to leave. The dual attribution of Theorem 3 charges each site the carbon
878 shadow price of its demand and credits it for the capacity it contributes. It is
879 budget-balanced, no coalition would prefer to secede, and it is a by-product
880 of the scheduling problem. It suits internal carbon pricing and the sharing
881 of the benefits of carbon-aware scheduling among business units. It is not
882 a substitute for external GHG Protocol reporting, where negative Scope 2
883 emissions do not exist [66]. The players must also be entities that could
884 credibly schedule on their own, such as separate operators, business units
885 with their own capacity, or tenants with reserved shares. Because credits
886 reward contributed capacity, capacity declarations should be audited.

887 *Deployment.* CARMA needs a carbon-intensity feed (average or marginal), a
888 demand profile computed from the operator’s own job logs and a network-flow
889 solver. It decides within tens of milliseconds and returns a plan that is optimal

890 for its model, which makes its decisions auditable. Absolute savings scale
 891 with the fleet: for the 1,700-server fleet simulated here, CARMA saves about
 892 16 tCO₂ per year relative to myopic control and about 165 tCO₂ relative to
 893 carbon-agnostic operation, in attributional terms.

894 6.2. Limitations and future research

895 The study has several limitations. First, the synthetic workloads use a
 896 profile learned from the same generator, which favours anticipation. The trace
 897 experiment removes this advantage, but it covers only four evaluation days of
 898 one cluster, and its deadlines and origins are synthetic. Second, the regional
 899 intensities published by NESO are modelled estimates, used here as realised
 900 values, and the marginal analysis relies on a stylised proxy, not on a dispatch
 901 model. Third, the facility parameters (PUE, network energy, a 96-core server)
 902 are stylised, and embodied emissions are outside the scope except as the floor
 903 that makes θ finite. Fourth, the forecaster excludes weather inputs, so its
 904 day-ahead skill is limited. Fifth, bag-of-tasks jobs with free preemption and
 905 migration are an idealisation: jobs with state, data gravity, bandwidth limits
 906 or data-sovereignty constraints would add restrictions that break the pure
 907 network structure. Sixth, the guarantees of Theorem 2 assume a backstop
 908 resource, a positive emission floor and an episode within the planning window,
 909 none of which holds in the simulations. The theorem describes the mechanism
 910 and the experiments its empirical profile. Its robustness term is shared by all
 911 on-time policies and is far from the $\Omega(\sqrt{\theta})$ lower bound. Seventh, the core
 912 property is exact only for the hindsight benchmark.

913 Several extensions follow naturally. The ghost-demand profile could be
 914 replaced by probabilistic arrival forecasts, with the anticipation weight set
 915 by a newsvendor-type argument, or by scenario-based planning that treats
 916 demand and carbon uncertainty jointly [57]. Tightening robustness towards
 917 the lower bound while retaining consistency, for example by protecting deferral
 918 decisions with reservation prices [24, 26], is an open theoretical question. The
 919 framework can also be driven by marginal emission signals, coupled with
 920 right-sizing, on-site renewables and storage [21, 22] and demand-response
 921 programmes [23], and evaluated across several electricity markets.

922 7. Conclusions

923 This paper studied carbon-aware scheduling of deferrable computing work-
 924 loads across data-centre sites as a sustainable-operations problem. We showed

925 that the offline problem, including width limits, deadlines and the emissions
 926 of data transfer, is a minimum-cost flow, which yields exact oracles and fast
 927 online planning. We proved that uncertainty about demand alone forces
 928 every online algorithm to a competitive ratio of order $\sqrt{\theta}$, even with perfect
 929 carbon forecasts, and that myopic planning does far worse. We proposed
 930 CARMA, which adds ghost jobs for expected arrivals to the planning network.
 931 Given backstop capacity, it meets every deadline under any predictions, is
 932 optimal under exact predictions and degrades linearly with prediction error.
 933 The dual prices of the pooled problem yield a carbon attribution that is
 934 budget-balanced and in the core of the pooling game.

935 In 52 out-of-sample weeks of 2024 on British regional grid data, with
 936 migration allowed, CARMA cut attributional emissions relative to carbon-
 937 agnostic operation by 36–68%. It came within 8.0–9.8% of the clairvoyant
 938 optimum, against 15–16% for the best of the myopic, protection-level and
 939 risk-averse baselines, and it did so in almost every week. On a real cluster
 940 trace, with a demand profile biased by 39% and bursty arrivals, its gap was
 941 20.7% against 26.7% for the best baseline. Anticipating demand was worth
 942 far more than perfect carbon foresight, and correlated day-level forecast errors
 943 hurt more than independent ones. Three results qualify these gains. Without
 944 migration, reductions are small (9–18%), and near saturation carbon-aware
 945 shifting delays work enough that carbon-agnostic operation is preferable under
 946 a service-adjusted metric. Under a stylised marginal-emission accounting, the
 947 achievable reduction fell to 14% and CARMA’s advantage over MPC was no
 948 longer significant, so the physical value of shifting within one grid depends
 949 on absorbing otherwise-curtailed low-carbon output. With always-on servers,
 950 most emissions no longer depend on the schedule.

951 Origin-based and host-based carbon accounting violated coalitional stabil-
 952 ity in every test week. The dual attribution never did in hindsight and failed
 953 6 of 728 checks, by at most 0.06 tCO₂, for CARMA’s realised emissions. Its
 954 charges were close to the Shapley value, which it obtains from a single solve.
 955 Carbon-aware computing is most effective when schedulers plan for the work
 956 that has not yet arrived, not only for the grid conditions that lie ahead, and
 957 when its benefits are measured and shared on a sound basis.

958 CRediT authorship contribution statement

959 **Quang-Vinh Dang:** Conceptualization, Methodology, Software, Data
 960 curation, Formal analysis, Investigation, Validation, Visualization, Writing –

961 original draft, Writing – review & editing.

962 Declaration of competing interest

963 The author declares no known competing financial interests or personal
964 relationships that could have appeared to influence the work reported in this
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969 Data availability

970 The carbon-intensity data are publicly available from the NESO Carbon
971 Intensity API [45] under the CC BY 4.0 licence. The data snapshot used in
972 this study, all code, the workload generator with fixed random seeds, and all
973 raw experiment outputs are available from the author’s repository (<https://github.com/vinhq dang/Sustainable-Operations-and-Computers>).
974

975 Appendix A. Proofs

976 *Atoms.* Splitting a job j into sub-jobs $j_1, \dots, j_k$ with work w_{j_i} and widths
977 $m_{j_i} = m_j w_{j_i} / w_j$ does not change the optimal value of any planning problem.
978 Given a solution for the merged job, the allocation $x_{j_i s}(t) = x_{j s}(t) w_{j_i} / w_j$
979 satisfies every constraint of the split problem at the same cost. Conversely,
980 summing the sub-job allocations gives a feasible allocation for the merged
981 job, because $\sum_i m_{j_i} = m_j$. Splitting also preserves window-feasibility, since
982 $w_{j_i} / m_{j_i} = w_j / m_j$. We may therefore compare true and predicted job sets
983 atom by atom.

984 Appendix A.1. Proof of Theorem 1

985 Normalise $L = 1$, so $U = \theta$. Site A has one server with cost $p \in [1, \theta]$ in
986 hour 0 and cost 1 in hour 1. Site B has ample capacity at cost θ in both
987 hours. Transfer costs are zero. A flexible job ($w = m = 1$, window $\{0, 1\}$) is
988 released at $t = 0$. The adversary either releases nothing more (scenario 1) or
989 releases an urgent job ($w = m = 1$, window $\{1\}$) at $t = 1$ (scenario 2). All
990 costs are known in advance.

Consider any deterministic online algorithm that may split work fractionally, and let $x \in [0, 1]$ be the amount of flexible work it runs at $t = 0$. Running at B in hour 0 costs $\theta \geq p$ and is dominated, so we may assume this work runs at A . In scenario 1 the algorithm finishes the remaining $1 - x$ at A in hour 1, so $\text{ALG}_1 = xp + (1 - x)$, while $\text{OPT}_1 = 1$. In scenario 2, hour 1 must process $2 - x$ units and A can take only one of them, so $\text{ALG}_2 = xp + 1 + (1 - x)\theta$, while $\text{OPT}_2 = p + 1$. The ratio $r_1(x) = 1 + x(p - 1)$ increases in x , and the ratio $r_2(x) = (1 + \theta - x(\theta - p))/(p + 1)$ decreases in x . Hence $\min_x \max\{r_1, r_2\}$ is attained where they are equal, at $x^* = (\theta - p)/(p^2 - p - 1 + \theta) \in [0, 1]$, and its value is

$$1 + \frac{(p - 1)(\theta - p)}{p^2 - p - 1 + \theta}.$$

The adversary chooses p to maximise this value, which gives $R^*(\theta)$. The choice $p = \sqrt{\theta}$ gives the stated closed-form lower bound. For a randomised algorithm, the expected cost in each scenario is affine in x , so it equals the cost of the deterministic fractional algorithm that plays $\mathbb{E}[x]$, and the bound applies against an oblivious adversary. Myopic MPC sees only the flexible job at $t = 0$ and, for any $p > 1$, defers it entirely to the cheaper hour 1 ($x = 0$). Its ratio in scenario 2 is $(1 + \theta)/(p + 1)$, which tends to $(1 + \theta)/2$ as $p \rightarrow 1^+$, so its competitive ratio is at least $(1 + \theta)/2$. $\square$

Appendix A.2. Proof of Proposition 3 and Theorem 2

Proposition 3. A policy that completes all work W on time executes every server-hour at a unit cost of at most U , so its emissions are at most UW . Every server-hour in the oracle's solution costs at least L , and slack costs $\Pi > U \geq L$, so $\text{OPT} \geq LW$. $\square$

Deadline safety. We first show by induction that every pending real job j is window-feasible at the start of every hour: its remaining work satisfies $r_j \leq m_j(d_j - t)$. This holds at release by (A1). Suppose it holds at hour t , and suppose the plan of CARMA leaves some real work $u_j > 0$ unfinished. Because $r_j \leq m_j(d_j - t)$, some hour $\tau \in [t, d_j)$ carries less than m_j units of job j in the plan. Moving a small amount of u_j to the backstop in hour τ keeps the plan feasible, because the backstop has unlimited capacity, and it changes the cost by a negative amount, because the backstop costs at most $U < \Pi$. This contradicts optimality, so the plan completes all real work. After the first hour is executed, the rest of the plan is a feasible completion,

1024 so $r'_j \leq m_j(d_j - t - 1)$ and the induction step holds. No real job is ever late,
 1025 whatever the ghost jobs are, and θ -competitiveness follows from Proposition 3.

1026 *Telescoping.* Let σ_t be the state at the start of hour t : the remaining work of
 1027 all real jobs released by t . Let $V_t(\sigma, G; c)$ be the optimal value of (2)–(5) over
 1028 hours $t, \dots, T-1$ for the pending jobs σ and the future jobs G under costs c .
 1029 Write $\bar{V}_t = V_t(\sigma_t, F_t; c)$, so that $\bar{V}_0 = \text{OPT}$ and $\bar{V}_T = 0$. For a feasible hour- t
 1030 allocation x with cost $g_t(x)$ and successor state $\sigma^+(x)$, define

$$Q_t(x) = g_t(x) + V_{t+1}(\sigma^+(x), F_t; c), \quad \hat{Q}_t(x) = g_t(x) + V_{t+1}(\sigma^+(x), \hat{F}_t; \hat{c}_t).$$

1031 By dynamic programming over a deterministic future, $\min_x Q_t(x) = \bar{V}_t$, and
 1032 $Q_t(x_t) = g_t(x_t) + \bar{V}_{t+1}$ for the executed action x_t . Summing over t gives

$$\text{ALG} - \text{OPT} = \sum_{t=0}^{T-1} \delta_t, \quad \delta_t = Q_t(x_t) - \min_x Q_t(x) \geq 0.$$

1033 Ghost jobs are released at $t+1$ or later, the planning window reaches T
 1034 by (A2), and hour- t costs are observed. The planning problem solved by
 1035 CARMA at hour t is therefore exactly $\min_x \hat{Q}_t(x)$, and x_t is a minimiser.

1036 *One-step loss.* Let $x^* \in \arg \min Q_t$. Since $\hat{Q}_t(x_t) \leq \hat{Q}_t(x^*)$, we have $\delta_t \leq$
 1037 $\Delta(x_t) - \Delta(x^*)$, where $\Delta = Q_t - \hat{Q}_t$. Write $\Delta = D_1 + D_2$ with

$$\begin{aligned} D_1(x) &= V_{t+1}(\sigma^+(x), F_t; c) - V_{t+1}(\sigma^+(x), \hat{F}_t; c), \\ D_2(x) &= V_{t+1}(\sigma^+(x), \hat{F}_t; c) - V_{t+1}(\sigma^+(x), \hat{F}_t; \hat{c}_t). \end{aligned}$$

1038 *Cost error.* An optimal solution under $\hat{c}_t$ is feasible under c , and its cost
 1039 changes by at most ε_t per unit of flow on future-hour arcs. The same holds
 1040 with the roles of c and $\hat{c}_t$ exchanged. Hence $|D_2(x)| \leq \varepsilon_t W_t$ for all x , and
 1041 $D_2(x_t) - D_2(x^*) \leq 2\varepsilon_t W_t$.

1042 *Demand error.* Under (A1), the penalty attached to a window-feasible
 1043 job does not affect any optimal value, provided it is at least U . If a solution
 1044 leaves slack on such a job, the argument used for robustness moves that slack
 1045 to the backstop at no greater cost. True future jobs (penalty Π) and ghost
 1046 jobs (penalty Π_g) can therefore be treated as identical atoms whenever they
 1047 coincide. Let a be a window-feasible atom and G any set of future jobs. For
 1048 every state σ ,

$$Lw_a \leq V(\sigma, G \cup \{a\}) - V(\sigma, G) \leq Uw_a.$$

1049 For the upper bound, add a to an optimal solution for G by running it on the
1050 backstop at rate $w_a/(d_a - a_a) \leq m_a$ in every hour of its window. For the lower
1051 bound, delete a 's flow and slack from an optimal solution for $G \cup \{a\}$; each
1052 deleted unit cost at least $\min\{L, \Pi_a\} = L$, since $\Pi_a \geq U \geq L$. Now transform
1053 $\hat{F}_t$ into F_t by removing $\hat{F}_t \setminus F_t$ and adding $F_t \setminus \hat{F}_t$, one atom at a time. Then
1054 $D_1(x)$ is a sum of increments, each lying in $[Lw_a, Uw_a]$ (with a minus sign
1055 for removals) for every x . Therefore $D_1(x_t) - D_1(x^*) \leq (U - L) \sum_a w_a =$
1056 $(U - L) d_t$.

1057 Combining the two bounds gives $\delta_t \leq (U - L)d_t + 2\varepsilon_t W_t$. Summing over t
1058 and taking the minimum with the robustness bound proves (9). If $d_t = \varepsilon_t = 0$
1059 for all t , then $\delta_t = 0$ and $\text{ALG} = \text{OPT}$. $\square$

1060 *Appendix A.3. Proof of Theorem 3*

1061 Write the problem of coalition C as $c(C) = \min\{c^\top z : A_- z =$
1062 $b_=(C), A_\leq z \leq b_\leq(C), z \geq 0\}$, where $z = (x, u)$. Jobs submitted out-
1063 side C have zero work and zero width, and sites outside C have zero capacity.
1064 This forces the corresponding flows to zero, so the problem coincides with
1065 the stand-alone problem of C . The right-hand side is additive over sites:
1066 $b(C) = \sum_{o \in C} b^o$, where b^o collects the work and widths of the jobs submitted
1067 at o and the capacity of site o . The matrix A and the cost c do not depend
1068 on C . The dual problem $\max\{b(C)^\top y : A^\top y \leq c, y_\leq \leq 0\}$ therefore has a
1069 feasible set that is the same for all coalitions.

1070 Let $y^* = (\lambda, \nu, \mu)$ be an optimal dual solution for $\mathcal{S}$. The slack variables
1071 make every primal problem feasible, and costs are non-negative, so strong
1072 duality gives $c(\mathcal{S}) = b(\mathcal{S})^\top y^* = \sum_o (b^o)^\top y^* = \sum_o \phi_o$, which is budget balance.
1073 For any coalition C , y^* is dual-feasible for C , so weak duality gives $c(C) \geq$
1074 $b(C)^\top y^* = \sum_{o \in C} \phi_o$, which is the core condition [29]. Expanding $(b^o)^\top y^*$
1075 gives (10). By Proposition 1 the problem is a min-cost flow. Its optimal node
1076 potentials, together with the reduced costs they induce on saturated capacity
1077 arcs, give an optimal dual solution, so network simplex returns y^* together
1078 with the optimal flow. $\square$

1079 **Appendix B. Forecast evaluation**

1080 Table B.6 and Fig. B.9a summarise forecast accuracy in 2024. Averaged
1081 over horizons and sites, the direct GBM forecaster has a mean absolute
1082 error (MAE) of 31.2 gCO₂/kWh, against 34.4 for persistence and 40.4 for
1083 the seasonal-naive forecast. The advantage is largest at short and medium

Table B.6: Forecast accuracy on the 2024 test year (MAE in gCO₂/kWh; mean over sites unless stated).

Model	Horizon h (hours)					All horizons	
	1	3	6	12	24	MAE	RMSE
Persistence	8.0	19.3	30.2	37.4	40.4	34.4	48.4
Seasonal naive (24 h)	40.4	40.4	40.4	40.4	40.4	40.4	57.0
Direct GBM (this study)	6.2	14.9	23.3	33.9	42.4	31.2	41.0

MAE by site over all horizons (persistence / GBM): South Scotland 14.1 / 16.2; North West England 24.7 / 21.8; West Midlands 56.1 / 49.3; London 42.7 / 37.6.

horizons: MAE falls by 23% at 1 hour (6.2 against 8.0) and by 23% at 6 hours (23.3 against 30.2). Beyond about 18 hours, all methods converge to the seasonal-naïve level, and at 24 hours the GBM is slightly worse than the seasonal-naïve benchmark. Without weather forecasts, day-ahead changes in wind output are hard to predict from the intensity history alone [44]. By site, the GBM is most accurate for the volatile southern sites. In South Scotland, where intensity is low and often flat, persistence is marginally better. Diebold–Mariano tests [68] with absolute-error loss and HAC variance confirm these patterns. For the three English sites, the GBM is significantly more accurate than persistence at horizons of 1–18 hours ($p \leq 0.001$) and indistinguishable at 24 hours ($p \geq 0.13$). For South Scotland it is significantly better at 1 hour, indistinguishable at 3 hours and significantly worse from 6 hours onwards.

Calibration matters more than point accuracy for the risk-adjusted cost view, and Fig. B.9b shows the effect of distribution shift. Static split-conformal bounds, calibrated on July–September 2023, under-cover in 2024 at every level: at $\beta = 0.9$, annual coverage was 0.858 and monthly coverage ranged from 0.68 to 0.95. With ACI, annual coverage matches the target to within 0.002 at all levels (0.500, 0.599, 0.698, 0.798 and 0.898). Monthly coverage stays within ± 4 percentage points of the target, and coverage is uniform across horizons. At $\beta = 0.5$, the ACI-adjusted median lies on average 9.9 gCO₂/kWh below the point forecast, which reflects the downward drift in intensity that the static model does not capture. Coverage is not the whole story. In pinball loss, ACI improves on static calibration at levels 0.5–0.7 (for example 14.83 against 14.90 at 0.5) but is worse at 0.8 and 0.9 (9.06 against 8.35 at 0.9). The online recalibration buys coverage at high levels with wider bounds. Two implementation details qualify the ACI results. Feedback is delayed: the

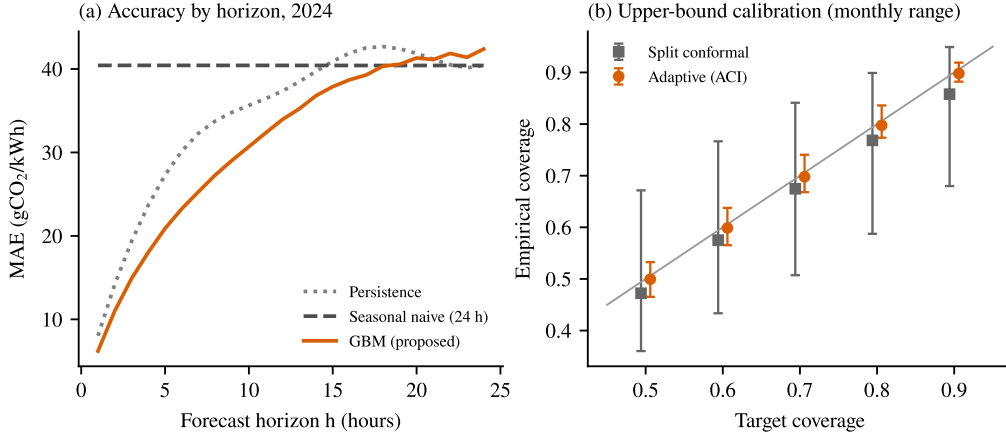


Figure B.9: Forecast evaluation in 2024. (a) MAE by horizon, averaged over sites. (b) Empirical coverage of one-sided upper bounds U^β against the target level β for static split conformal calibration and adaptive conformal inference (ACI). Markers show the annual coverage and bars the range across the 12 months.

1110 miscoverage level for horizon h is updated only when the outcome is observed,
 1111 h hours after issue, so the adaptation lags by up to a day at long horizons.
 1112 The adapted level is also clipped to $[0, 1]$ and read from a quantile grid of step
 1113 0.001, so the finite-sample guarantee of ACI [54] holds only approximately.

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